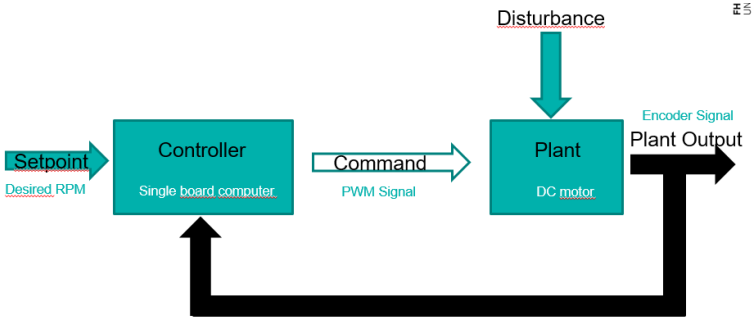
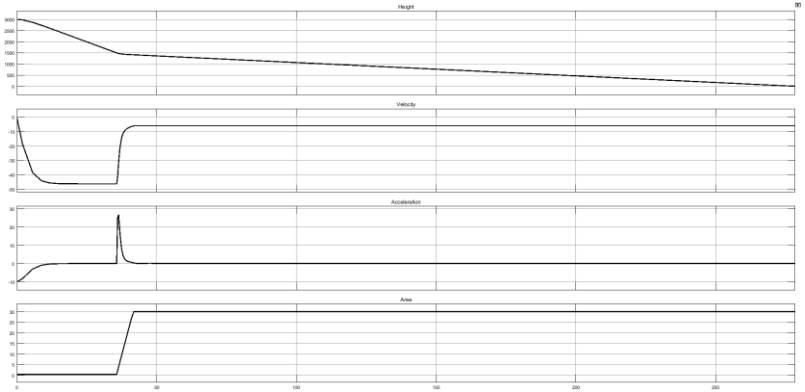
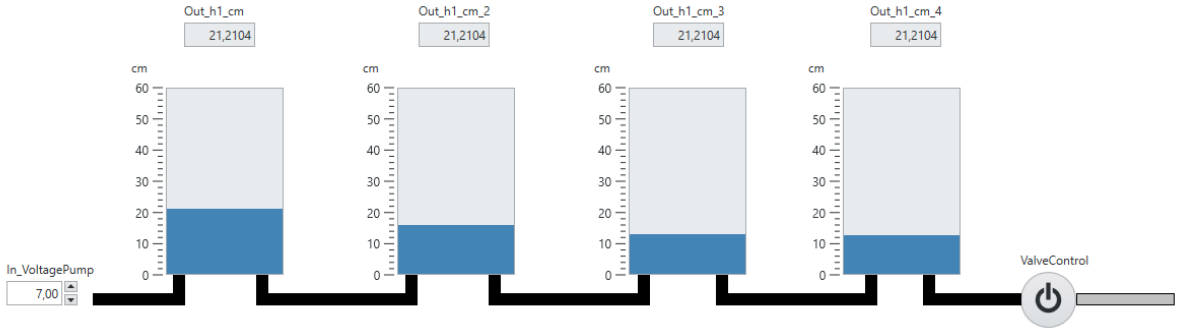
Mechatronic Systems Simulation

Model based design and component level test







Mechatronic Systems Simulation
What describes a model?

- A model is a piece of code or program which is developed by using specified tools (e.g. Excel, LabView, MATLAB…).
- A modeling is the process of abstraction of a real system.
- It can describe a system and can be viewed as an abstraction of an actual system or a physical replica of a system or a situation.
- The abstracted model may be logical or mathematical.
- A mathematical model is a mathematical description of properties and interactions of the system.
- The development of a model depends on the system boundary, system components, and their interactions (input and output).

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Mechatronic Systems Simulation
Where do you get a model?

Three primary ways to obtain a model:

- Easiest way: Find an existing model.
- For some system, it is possible to create a (theoretical) model based on knowledge of the system dynamics. In this option you need detailed knowledge of how the system works and might require domain expertise.
- Use test data to generate a model using system identification techniques (e.g. step- response). This techniques is useful if you have no knowledge of the system or if the system is too complex. This technique might be more accurate, because it is based on actual hardware behavior instead of theoretical calculations.

Mechatronic Systems Simulation
Model development

Validation of a model results

- Plausibility check and qualitative analysis
- Quantitative analysis including border analysis
- Compare results against the real data
- Compare results against simple models
- Document results and compare it

Mechatronic System Simulation

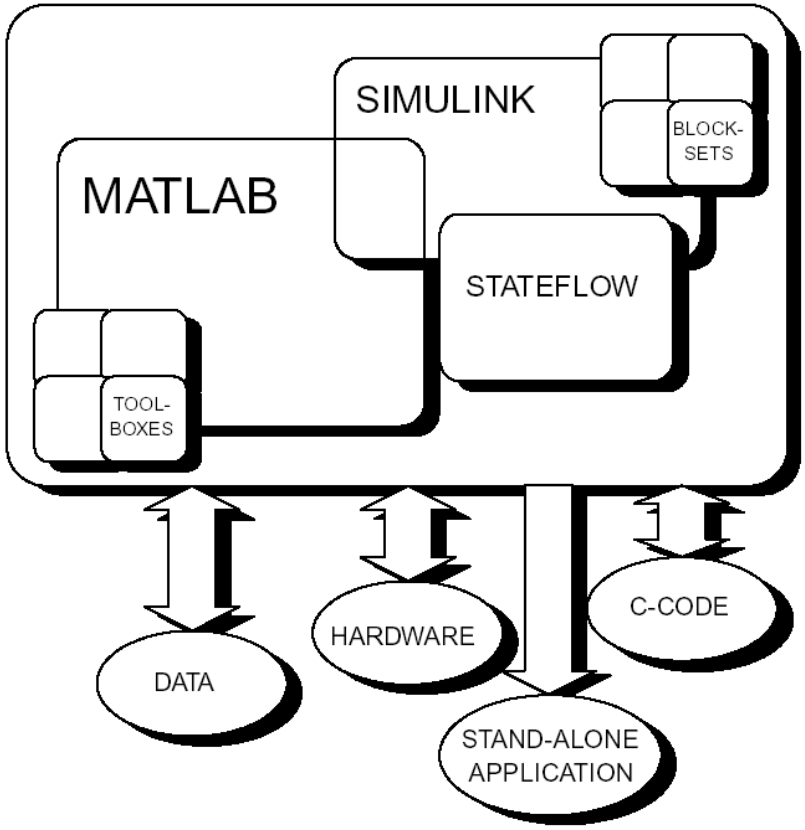
Seminar Overview: Contents

Part 1: Code based models

- Introduction
- MATLAB interactive
- MATLAB as a programming language
- MATLAB Graphics
- MATLAB functions (built in)
- MATLAB Data Interfaces (In-/ Output)

Mechatronic System Simulation

Introduction



What is MATLAB?

MATLAB is the abbreviated form of **Matrix Laboratory**.
It has been developed for technical Computing.

Key features

- Interactive programming environment for engineers and scientists
- High level language for technical computing
- Mathematical function libraries
- Graphics and data visualization tools
- External Interfaces: C, C++, Fortran, Excel, .Net
- Release of MATLAB 1.0 in 1984 (commercial), as university tool since early 70s

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Mechatronic System Simulation

Seminar Overview: Contents

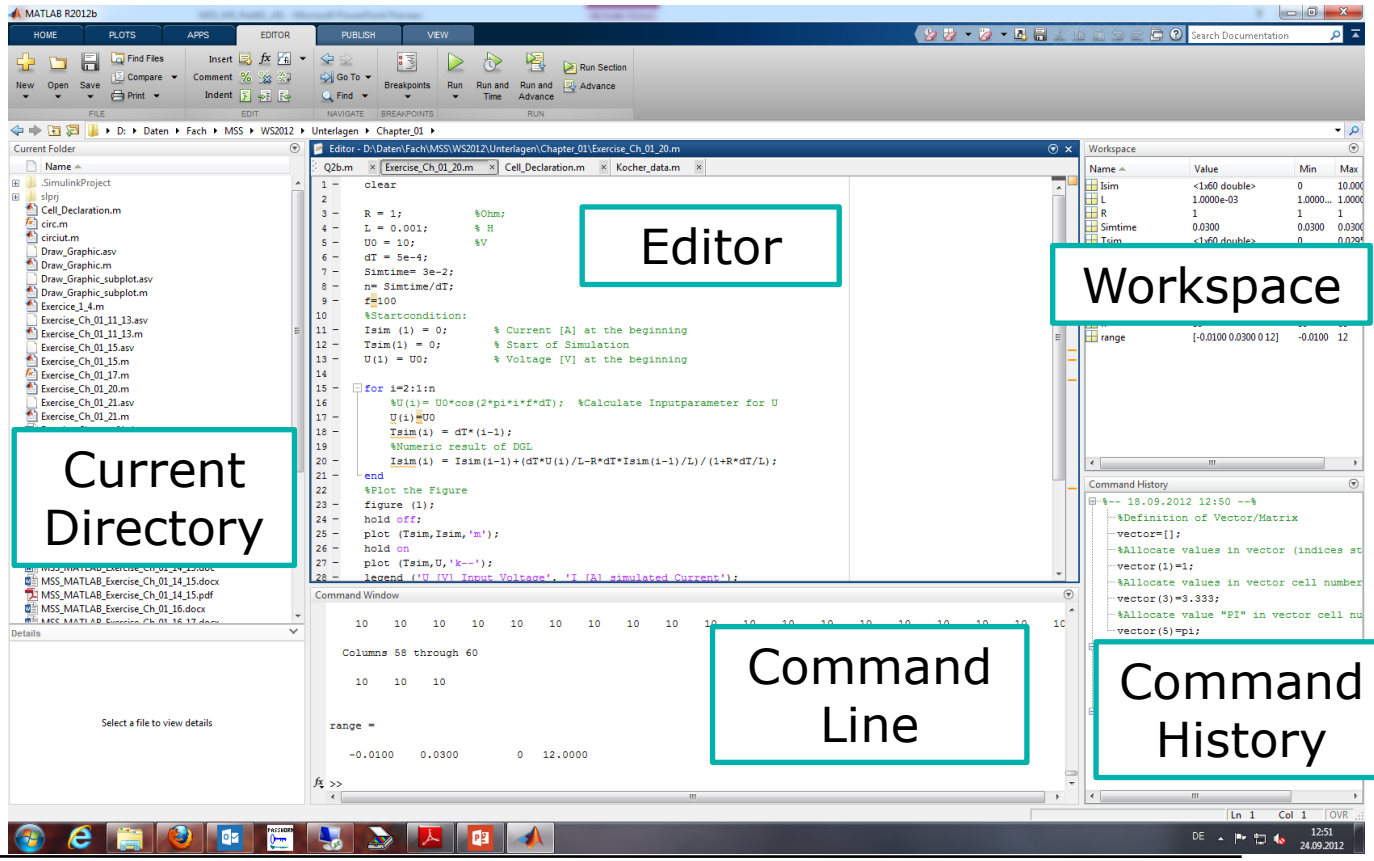
Part 1: Code based models

- Introduction
- MATLAB interactive
- MATLAB as a programming language
- MATLAB Graphics
- MATLAB functions (built in)
- MATLAB Data Interfaces (In-/ Output)

Mechatronic System Simulation
MATLAB Command Desktop

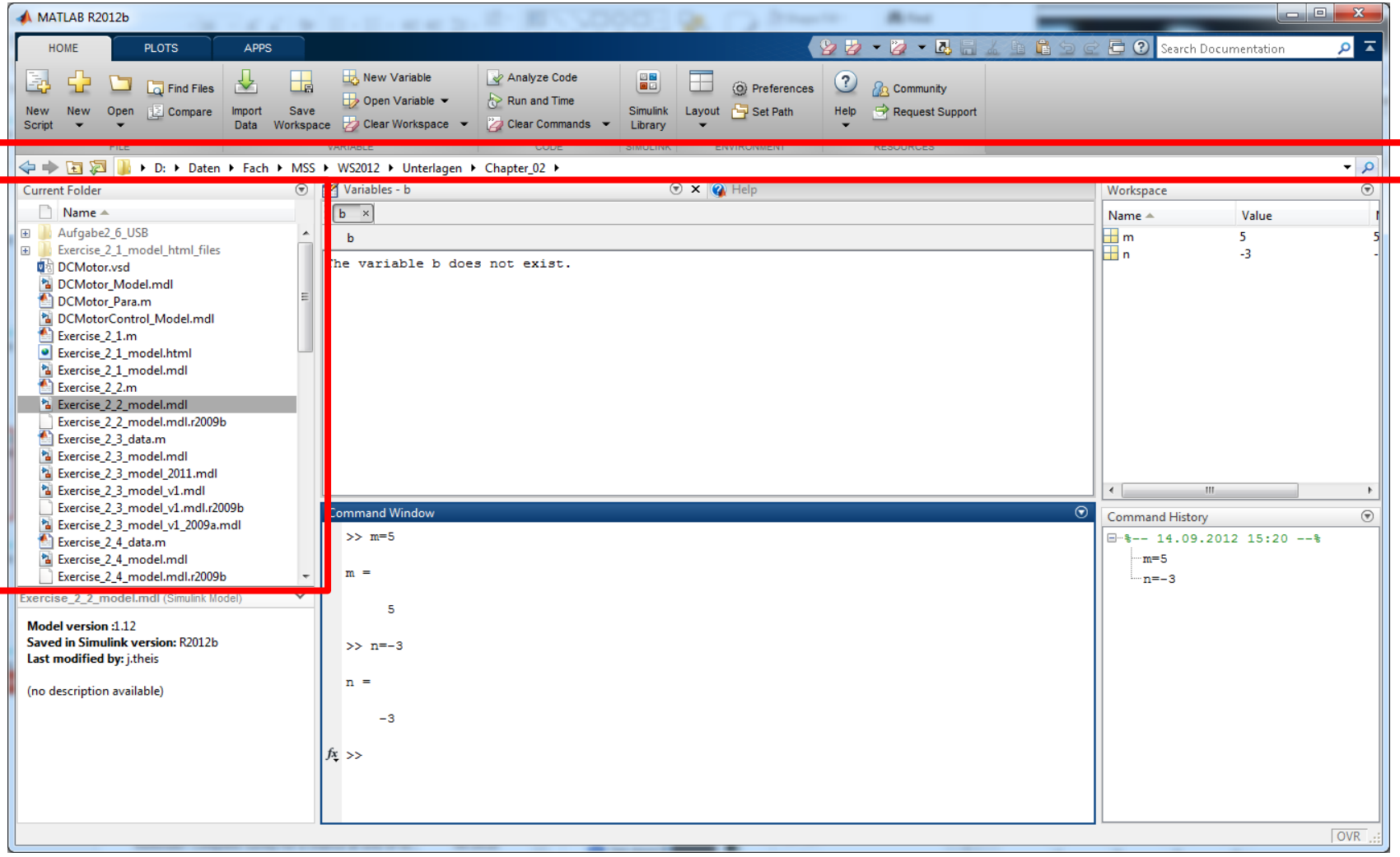
Integrated Development Environment

- Command Line
- Editor
- Command History
- Workspace
- Current Directory

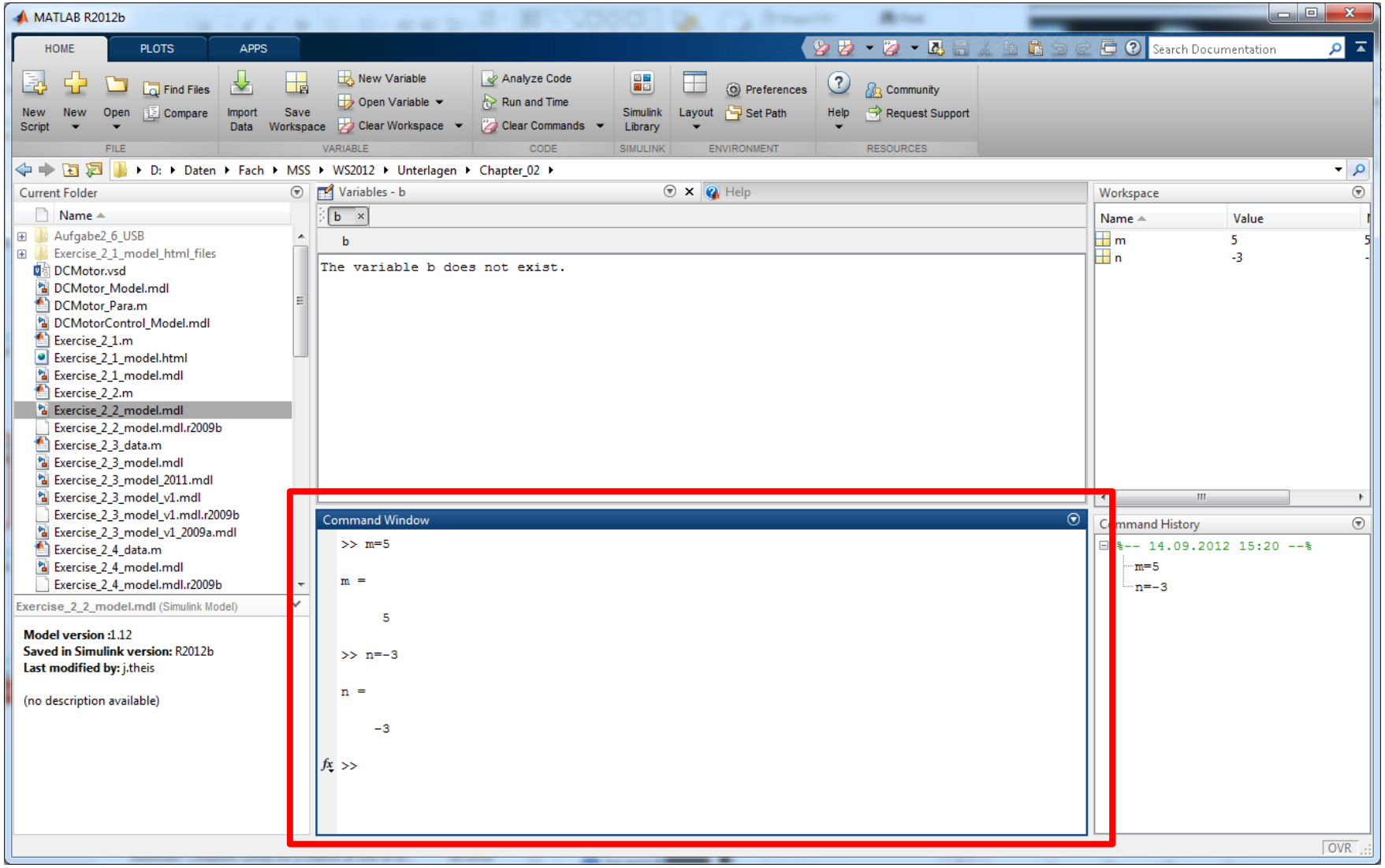


Mechatronic System Simulation
MATLAB: Current Folder

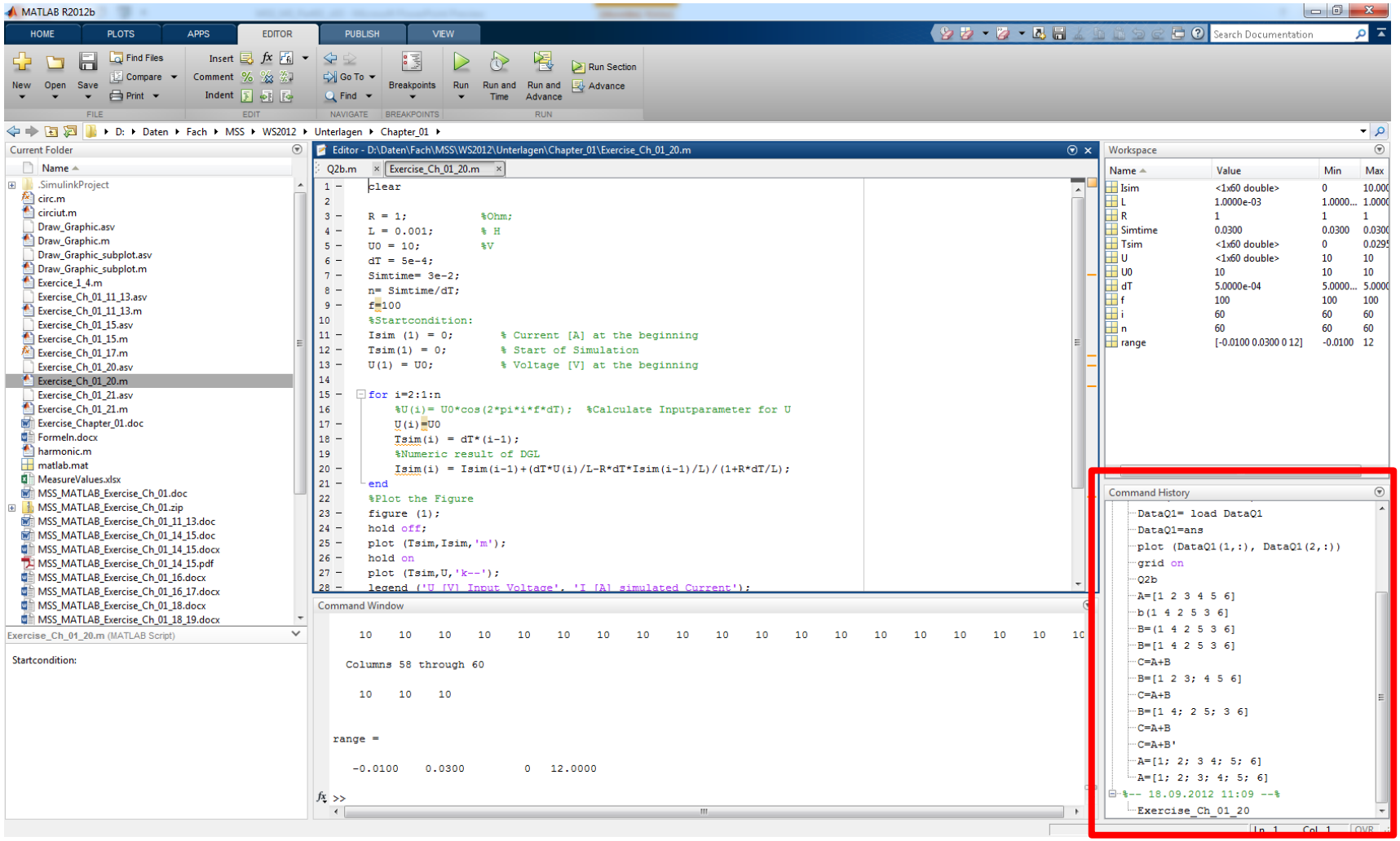
Define your current folder to the local drive D:/



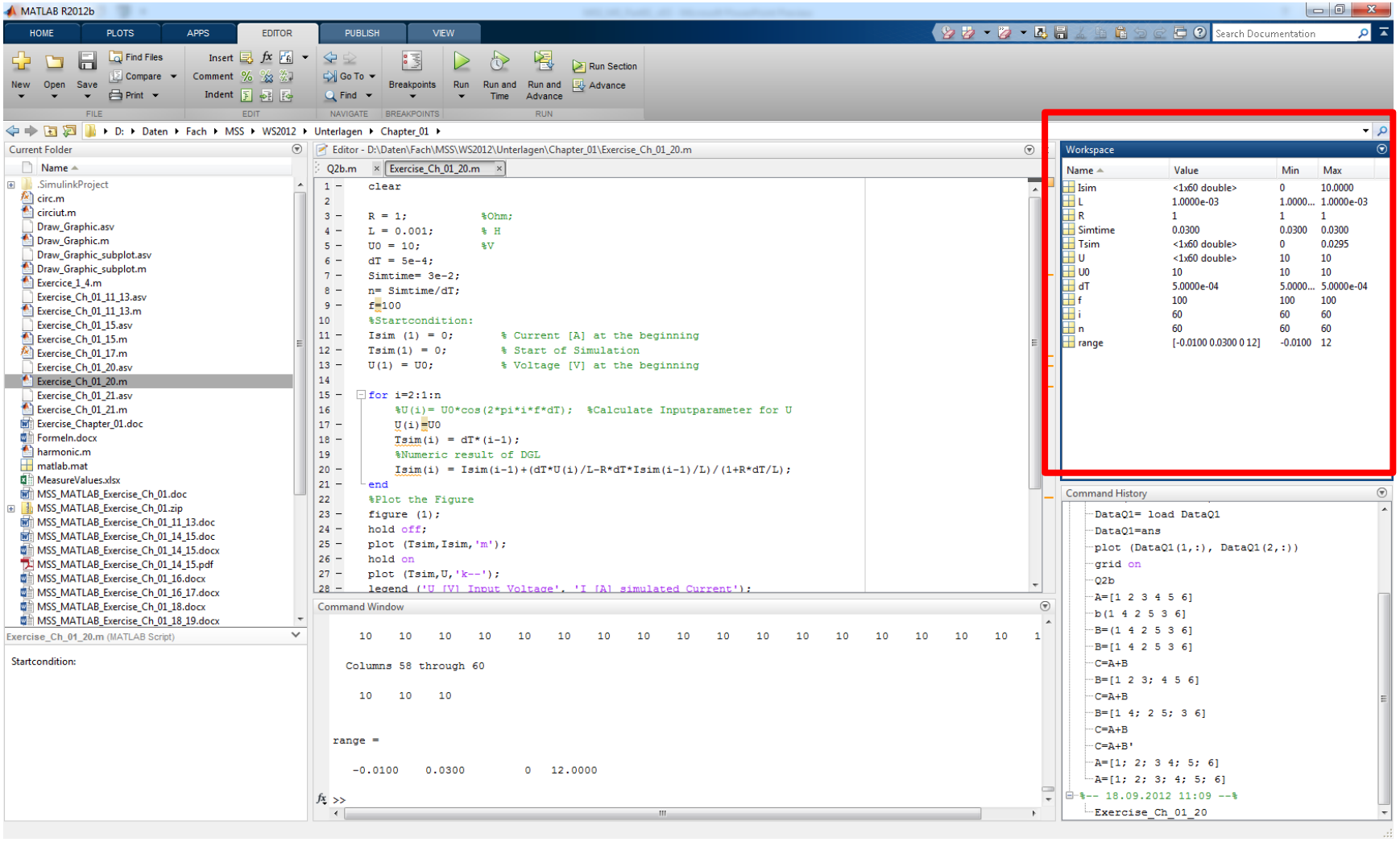
Mechatronic System Simulation
MATLAB: Command Window



Mechatronic System Simulation
MATLAB: Command History



Mechatronic System Simulation
MATLAB: Workspace



Mechatronic System Simulation
MATLAB: Notation

Industry Standards

MAAB- The MathWorks Automotive Advisory Board

Established in 1998 by MathWorks, Ford, Daimler-Benz and Toyota

- Modeling style guidelines for MathWorks controls, simulation, code generation products including Simulink, Stateflow and Embedded Coder.

CONTROL ALGORITHM MODELING GUIDELINES USING MATLAB®, Simulink®, and Stateflow®
Version 3.0

MathWorks Automotive Advisory Board (MAAB)

Guide can be find on ILIAS (MSS folders)

4.Naming Conventions

4.1. General Guidelines

4.1.1. ar_0001: Filenames

ID: Title	**ar_0001: Filenames**
Priority	Mandatory
Scope	MAAB
MATLAB Version	All
Prerequisites	
Description	A filename conforms to the following constraints: FORM: filename = name.extension **name**: no leading digits, no blanks **extension**: no blanks UNIQUENESS: ☐ all filenames within the parent project directory; ☐ cannot conflict with C / C++ or MATLAB keywords ALLOWED CHARACTERS: **name** a b c d e f g h i j k l m n o p q r s t u v w x y z A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 0 1 2 3 4 5 6 7 8 9 _ **extension**: a b c d e f g h i j k l m n o p q r s t u v w x y z A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 0 1 2 3 4 5 6 7 8 9 UNDERSCORES: **name**: can use underscores to separate parts; cannot have more than one consecutive underscore; cannot start with an underscore; cannot end with an underscore. **extension**: should not use underscores
Rationale	☑ Readability ☑ Workflow ☑ Simulation ☐ Verification and Validation ☑ Code Generation
Last Change	V3.00

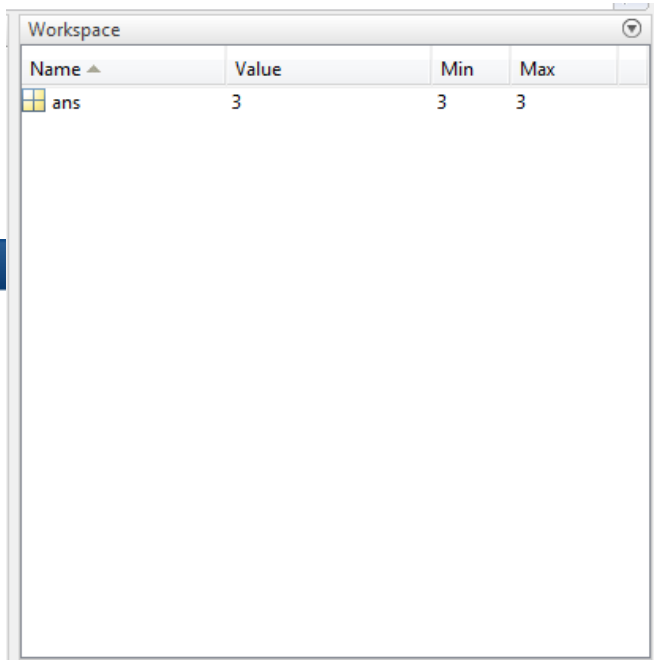
Mechatronic System Simulation
MATLAB: Notation of variables

Names:

- Maximal 63 characters (Including „_“, characters and numbers)
- First Character must be a letter!
- Capital and small letters are distinguished by MATLAB
- Variables are global declared in the workspace.
- Standard output answer variable: *ans*

Example calculation without variables)





Mechatronic System Simulation
MATLAB: Notation of variables

Example 2: (Simple calculation by using variables):

Command Window

```
>> A=1;
>> B=2;
>> C=A+B

C =

     3

>>
```

Workspace

Name	Value	Min	Max
A	1	1	1
B	2	2	2
C	3	3	3

Command History

```
%-- 18.09.2012 11:24 --%
1+2
A=1;
B=2;
C=A+B
A=1;
B=2;
C=A+B
A=1;
B=2;
C=A+B
A=1
A=1;
B=2;
C=A+B
```

Mechatronic System Simulation
MATLAB: Notation

[ ]	Enclose vectors and matrices Enclose output parameters of functions	`A = [ 1 2;3 4]` `[V,D]=eig(A);`

( )	Indices of matrix or vector elements
	Nesting of mathematical calculations
	Input parameters of functions
		`A(3);`
		`(a+b)*c`
		`plot(x,y)`

{ }	Representation of cells	`cell(1)={A};`
=	Assignment	`A=B;` `i=i+1;`

.	Decimal point
	Elementwise mathematical operation
	Structures
	Parent directory
	Continuation
		`1.2`
		`A.*B;`
		`struct.number=A`
		`cd ..`
		`a = b^2 ...`
		`+ 2*b*c + c^2`

,	Separation of matrix subscripts
	Separation of function arguments
	Multi statement lines
		`[1,2,3,4]`
		`zeros(2,5);`
		`if i==1,i, end`

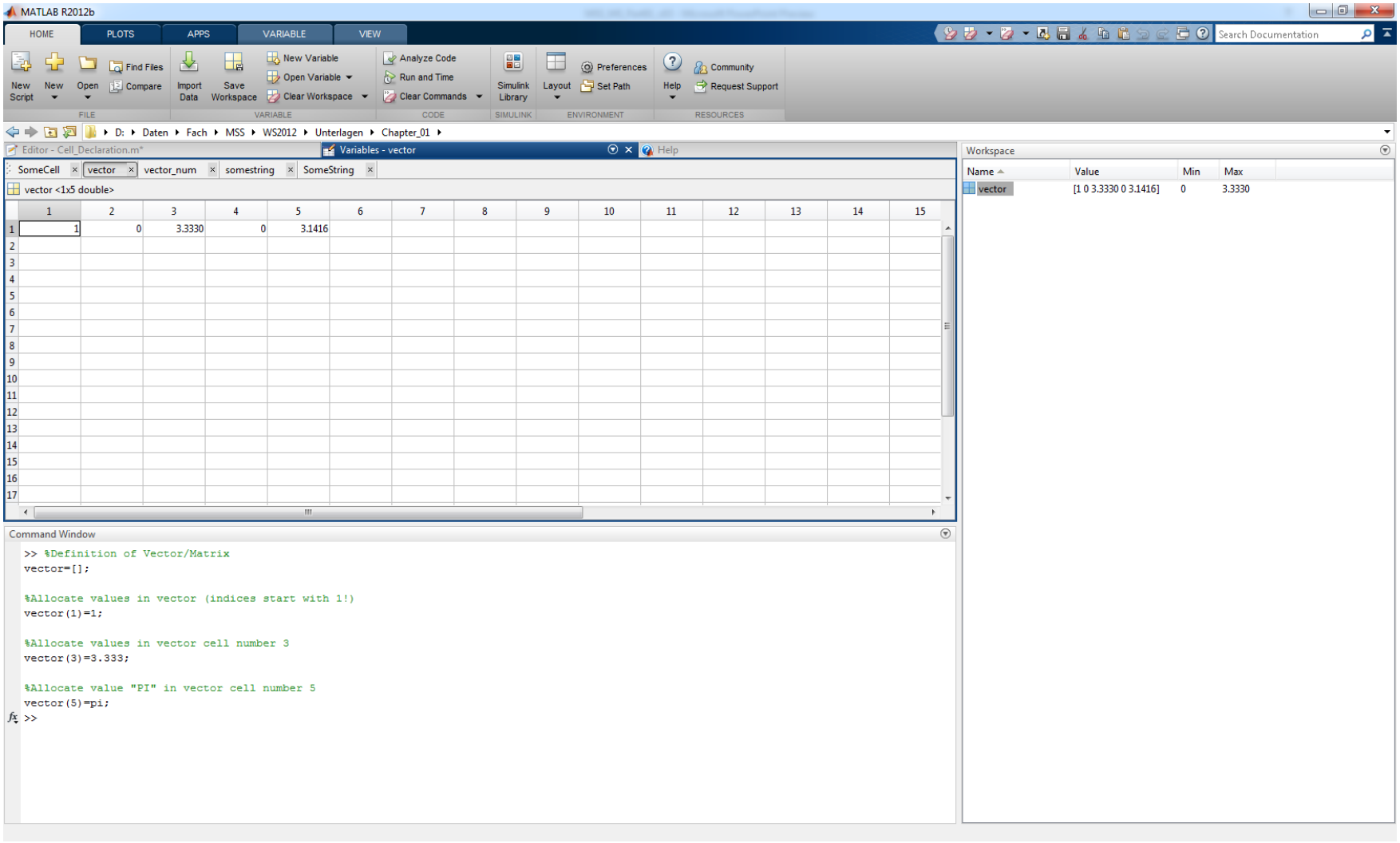
Mechatronic System Simulation
MATLAB: Notation

;	End of row in matrices Supresses the display of results	`[1;2;3;4]` `[1;2;3;4];`
%	Comment; rest of the line is not executed	`%Comment line`
!	Command to the operating system	`!dir`

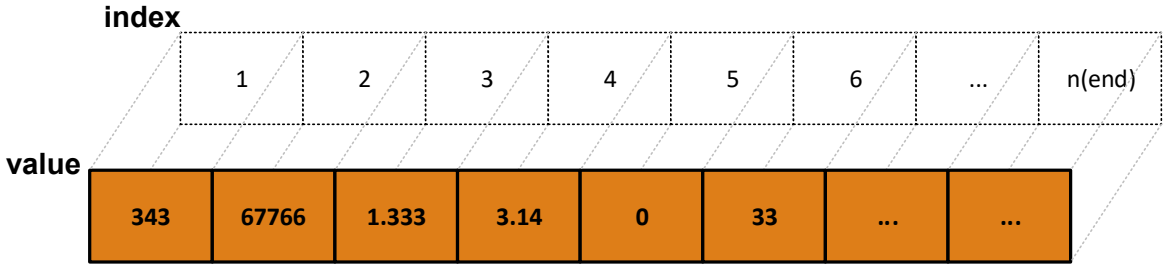
:	Generation of vectors
	Submatrices
	Loop variables
		`A=1:.1:5;`
		`A(:,2)`
		`for i=1:10,..`

'	Encloses strings Transpose of vectors and matrices	`string='Hello'` `B=A'`

Mechatronic System Simulation
MATLAB: Notation of Vector/Matrix: Allocate values



Mechatronic System Simulation
MATLAB: Notation of a Vector



Vector A (index)

First element has index 1!

Define Vector: A=[343 67766 1.333 3.14 0 33]

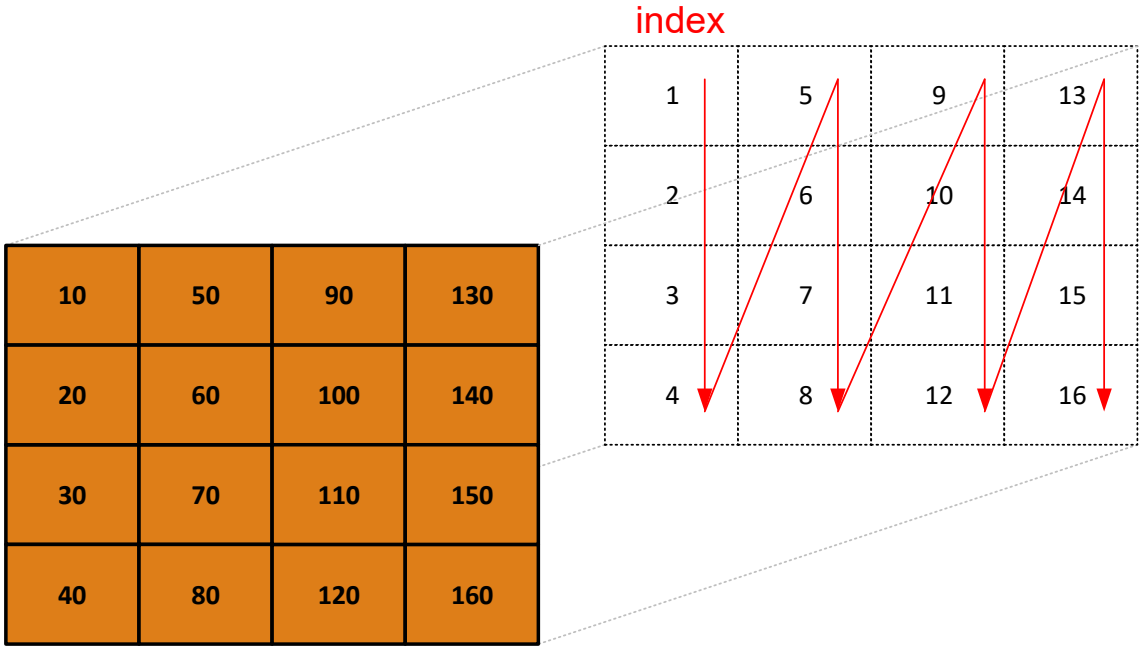
Define single Element: A(2)=67766

Index last element: „end“ B=A(end);

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Mechatronic System Simulation
MATLAB: Notation of a Matrix

Matrix A (row)



First element has index 1

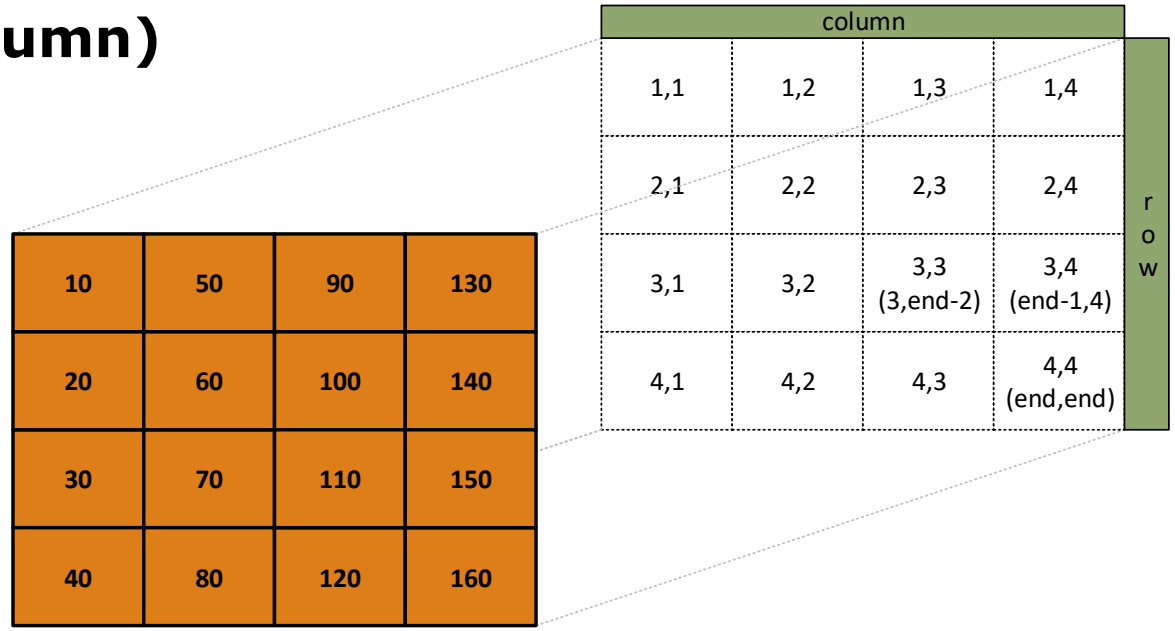
A=[10 50 90 130; 20 60 100 140; 30 70 110 150; 40 80 120 160]

Define single Element: A(3)=30

Index last elements: „end“ B=A(end); -> 160

Mechatronic System Simulation
MATLAB: Notation of a Matrix

Matrix A (row,column)



First element has Index 1,1!

A=[10 50 90 130; 20 60 100 140; 30 70 110 150; 40 80 120 160]

Define single Element: A(3,3)=110

Index last elements: „end“ B=A(end,end); -> 160

Mechatronic System Simulation
MATLAB: Assignment of vectors and matrices

$$A = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

- Elementwise assignment

```
>> A(1,1)=1; A(1,2)=4; A(1,3)=7;
>> A(2,1)=2; A(2,2)=5; A(2,3)=8;
>> A(3,1)=3; A(3,2)=6; A(3,3)=9;
```

- Using MATLAB notation

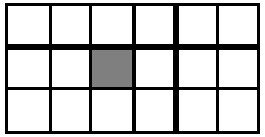
```
>> A=[1 4 7;2 5 8;3 6 9];
```

- Using operators or other functions

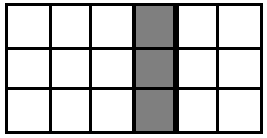
```
>> A=1:9; A=reshape(A,3,3);
```

Mechatronic System Simulation
MATLAB: Sub Matrices, Matrix- Indexing

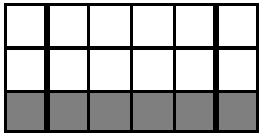
- Single Elements: A(2,3)



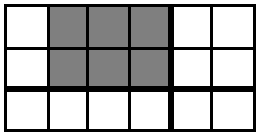
- Columns: A(:,4)



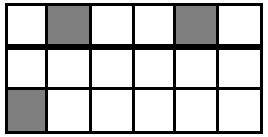
- Rows: A(3,:)



- Blocks: A (1:2,2:4)

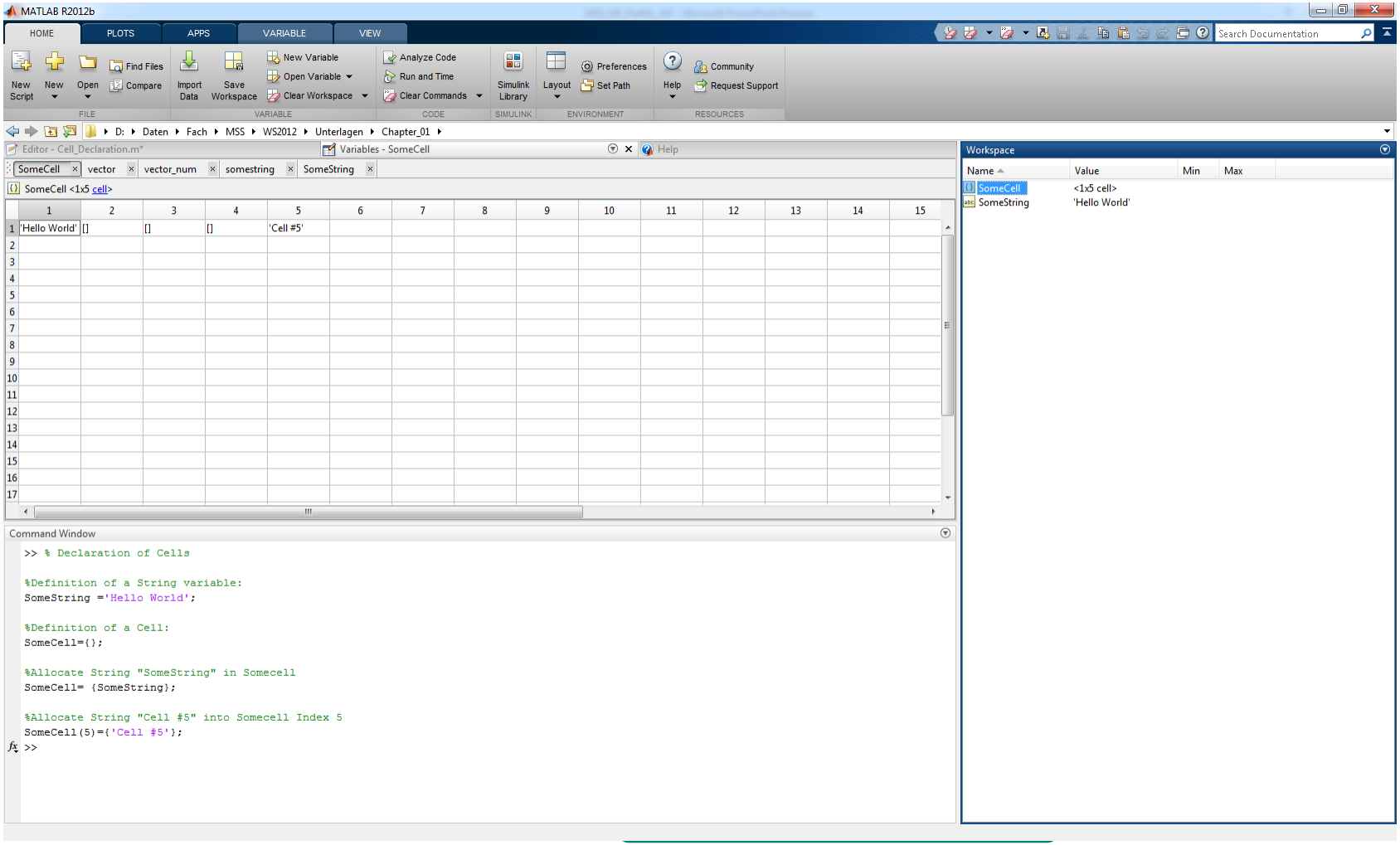


- Several Elements: A([3 4 13])



- First Element: Has index 1!
- Last Element: matrix(:,end)

Mechatronic System Simulation
MATLAB: Notation of Cells



Mechatronic System Simulation
MATLAB: Notation of Cells

Cell Array

- Store values of different types. Cell arrays can be a vector, matrix or string.
- Different values are stored in different elements of the array.
- Example: Store different strings of different lengths.

Matrix:

	1	2
1	23	'text'
2	[1 2 3 4 5 6 7 8 9 10]	'text'

Element of a vector inside a cell:

```
>> Cell_Matrix{2,1}(3)

ans =

     3
```

```
>> Cell_Matrix={23 'text'; (1:10) 'text'}

Cell_Matrix =

    [          23]    'text'
    [1x10 double]    'text'

>> Cell_Matrix(1)

ans =

    [23]

>> Cell_Matrix(2,1)

ans =

    [1x10 double]
```

Mechatronic System Simulation
MATLAB: Vector design

- Elements in identical step width: *A = (begin:step:end)*
  Example:

```
Command Window
>> A=(1:2:10)

A =

     1     3     5     7     9

fx >>
```

- Linear vector: *A= linspace (begin, end, number of elements)*

```
Command Window
>> A=linspace(1,10,5)

A =

    1.0000    3.2500    5.5000    7.7500   10.0000

fx >>
```

Mechatronic System Simulation
MATLAB: Vector design

- Logorithmic vector: *A= logspace(begin, end, number)*

```
Command Window
>> A=logspace(-2,2,5)

A =

    0.0100    0.1000    1.0000   10.0000  100.0000

fx >>
```

- Random vector:
  (Random 0 -1)

```
Command Window
>> A=rand(1,5)

A =

    0.7572    0.7537    0.3804    0.5678    0.0759

fx >>
```

Mechatronic System Simulation
MATLAB: Matrics Operators

	matrix operation	field operation
addition	+	+
subtraction	-	-
multiplication	*	.*
division right	/	./
division left	\	.\
power	^	.^
transpose (conj. complex)	'	
transpose	.'	

```
>> A=[1 2;3 4]
A=
     1     2
     3     4

>> B=A;
>> C=A*B

C =
     7    10
    15    22

>> D=A.*B

D =
     1     4
     9    16
```

Mechatronic System Simulation
MATLAB: Relational and logical operators

<	less than	a<b
<=	less than or equal	a<=b
>	greater than	a>b
>=	greater than or equal	a>=b
==	equal	a==b
&	and	a&b
|	or	a|b
~	not	~a
~=	not equal	a~=b

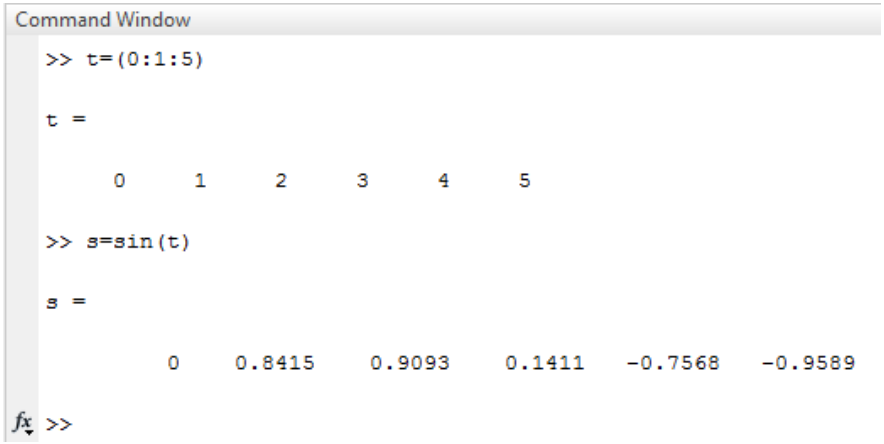
Mechatronic System Simulation
MATLAB: Elementary Mathematical Function

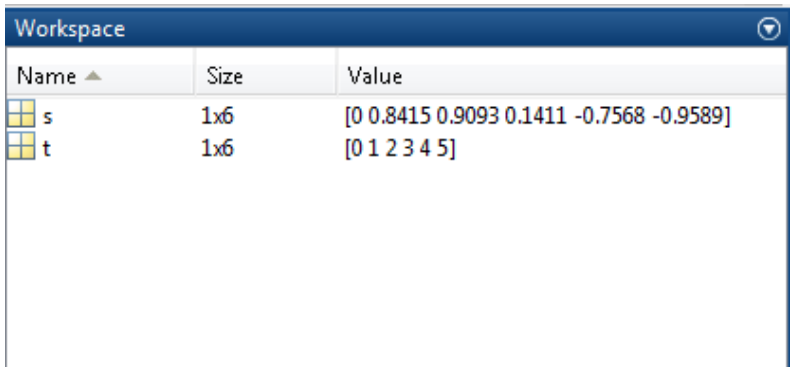
`abs`	Absolute Value
`angle`	Phase angle
`atan`	Inverse tangent
`ceil`	Round towards $+\infty$
`conj`	Complex conjugate
`cos`	Cosine
`exp`	Exponential
`fix`	Round towards zero
`floor`	Round towards $-\infty$
`imag`	Complex imaginary part

`log10`	Common (base 10) logarithm
`real`	Complex real part
`rem`	Remainder after division
`round`	Round towards nearest integer
`sign`	Signum function
`sin`	Sine
`sqrt`	Square root
`tan`	Tangent
`tanh`	Hyperbolic tangent

Mechatronic System Simulation
MATLAB: Mathematical functions

Example 3: (Simple calculation by Matlab notation):





Matlab notation replaced loop conditions.

C- Code:

```
double s[6]
for (i=0; i<6; i++)
s[i] = sin (i);
```

Mechatronic System Simulation
MATLAB: Special matrices

- Matrix of zeros

```
>> A=zeros(3,3) % or: A=zeros(3)
A =
     0     0     0
     0     0     0
     0     0     0
```

- Matrix of ones

```
>> A=ones(3,3)  % or: A=ones(3)
A =
     1     1     1
     1     1     1
     1     1     1
```

- Indentity matrix

```
>> A=eye(3,3)
A =
     1     0     0
     0     1     0
     0     0     1
```

- Empty matrix

```
>> A=[]
A =
     []
```

Mechatronic System Simulation
MATLAB: Administration of variables

- Dimension
  - Vector: >>length *(vector*)
  - Matrix: >>size (*matrix*)
- Display
  - who (variable) >>displays only names
  - who >>shows all names from workspace
  - whos(variable) >>displays name, size, bytes and class
  - whos >>displays all variables in details from workspace
- Delete
  - Delete variable: >>clear variable
  - Delete all variables in workspace: >>clear, clear all
  - Existent of variable >>exist variable
- Save/load variables to workspace: >>save [somestring] load[somestring]

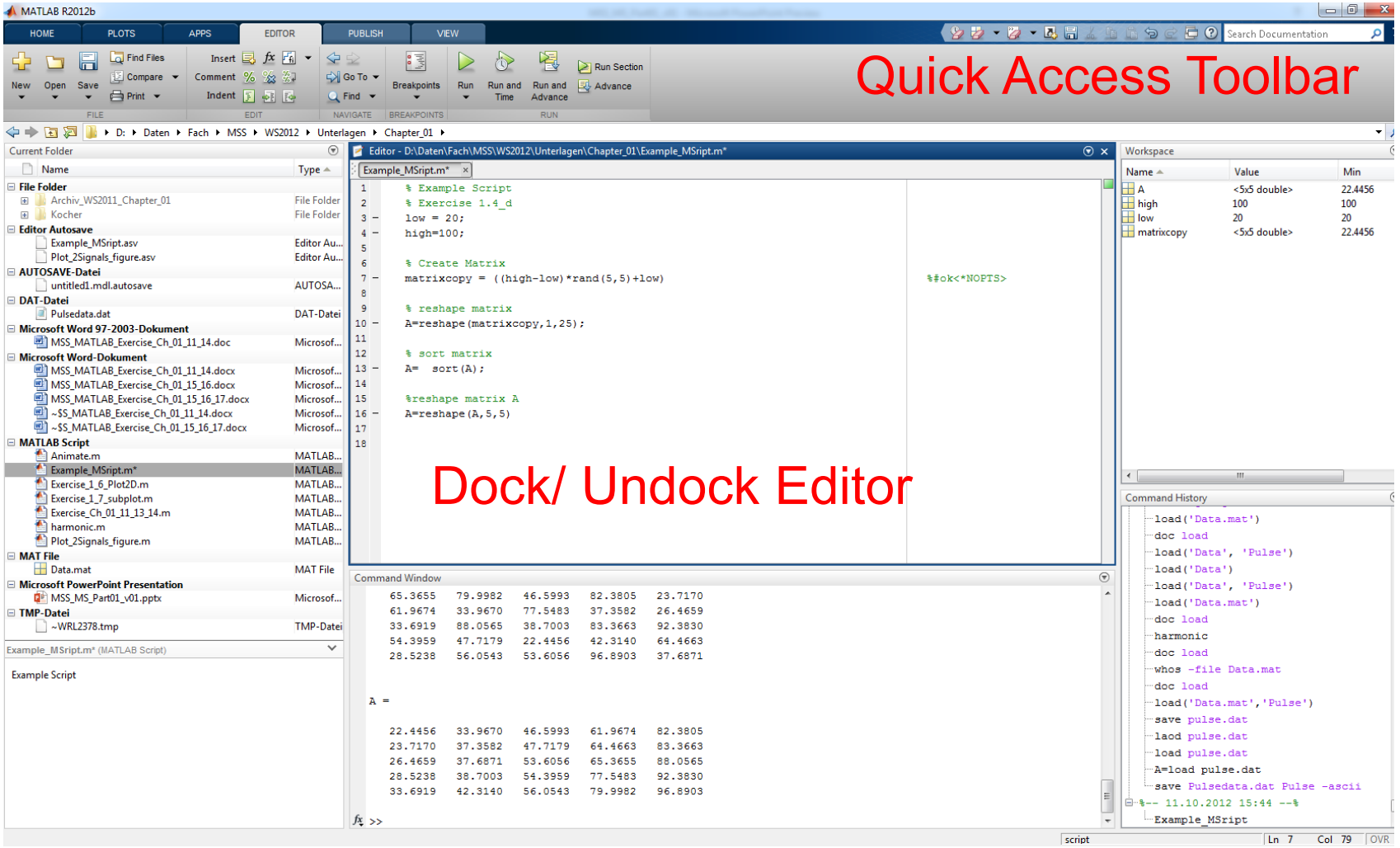
Mechatronic System Simulation

Seminar Overview: Contents

Part 1: Code based models

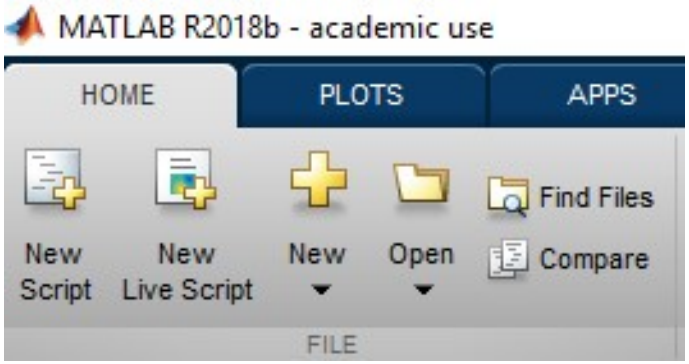
- Introduction
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- MATLAB Data Interfaces (In-/ Output)

Mechatronic System Simulation
Matlab Desktop (Layout and Quick Access Toolbar)

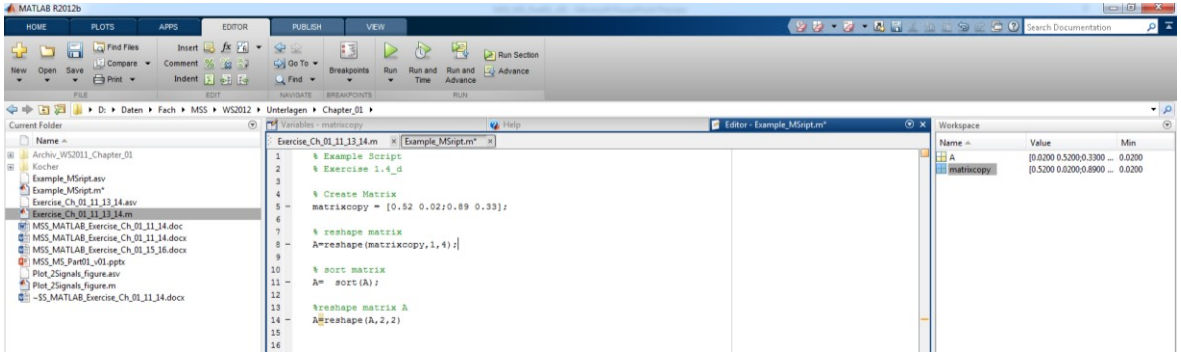


Mechatronic System Simulation
MATLAB Script

- Script: Commands in ASCII- File
  Format „Filenname.m“
  (instead of using the command line)



- Run the Script for debug and execution
  (e.g. Variables will be created in the workspace and files are shown in the current folder



Mechatronic System Simulation
MATLAB Script Example

Script:

```
%% Create Matrix
matrixrand=rand(5,5)
matrixcopy = matrixrand

% reshape matrix
A=reshape(matrixcopy,1,25);

% sort matrix
A=  sort(A);
Amin=min(reshape(matrixcopy,1,25))
Amax=max(reshape(A,1,25))

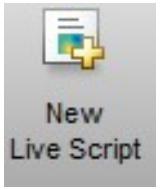
%reshape matrix A
A=reshape(A,5,5)
```

Output Command window

```
matrixrand =
    0.2760    0.4984    0.7513    0.9593    0.8407
    0.6797    0.9597    0.2551    0.5472    0.2543
    0.6551    0.3404    0.5060    0.1386    0.8143
    0.1626    0.5853    0.6991    0.1493    0.2435
    0.1190    0.2238    0.8909    0.2575    0.9293
matrixcopy =
    0.2760    0.4984    0.7513    0.9593    0.8407
    0.6797    0.9597    0.2551    0.5472    0.2543
    0.6551    0.3404    0.5060    0.1386    0.8143
    0.1626    0.5853    0.6991    0.1493    0.2435
    0.1190    0.2238    0.8909    0.2575    0.9293
Amin =
    0.1190
Amax =
    0.9597
A =
    0.1190    0.2435    0.3404    0.6551    0.8407
    0.1386    0.2543    0.4984    0.6797    0.8909
    0.1493    0.2551    0.5060    0.6991    0.9293
    0.1626    0.2575    0.5472    0.7513    0.9593
    0.2238    0.2760    0.5853    0.8143    0.9597
```

Code and results can not be in one file!

Mechatronic System Simulation
MATLAB Live Script Example



Exercise 1.4_d

```
% Create Matrix
matrixrand=rand(5,5)
matrixcopy = matrixrand

% reshape matrix
A=reshape(matrixcopy,1,25)

% sort matrix
A=  sort(A);
Amin=min(reshape(matrixcopy,1,25))
Amax=max(reshape(A,1,25))

%reshape matrix A
A=reshape(A,5,5)
```

```
matrixrand = 5×5
    0.7577    0.7060    0.8235    0.4387    0.4898
    0.7431    0.0318    0.6948    0.3816    0.4456
    0.3922    0.2769    0.3171    0.7655    0.6463
    0.6555    0.0462    0.9502    0.7952    0.7094
    0.1712    0.0971    0.0344    0.1869    0.7547

matrixcopy = 5×5
    0.7577    0.7060    0.8235    0.4387    0.4898
    0.7431    0.0318    0.6948    0.3816    0.4456
    0.3922    0.2769    0.3171    0.7655    0.6463
    0.6555    0.0462    0.9502    0.7952    0.7094
    0.1712    0.0971    0.0344    0.1869    0.7547

A = 1×25
    0.7577    0.7431    0.3922    0.6555    0.1712    0.7060 ...

Amin = 0.0318
Amax = 0.9502
A = 5×5
    0.0318    0.1869    0.4387    0.6948    0.7577
    0.0344    0.2769    0.4456    0.7060    0.7655
    0.0462    0.3171    0.4898    0.7094    0.7952
    0.0971    0.3816    0.6463    0.7431    0.8235
    0.1712    0.3922    0.6555    0.7547    0.9502
```

- Live scripts and live functions are program files useful for interacting with MATLAB commands.
- Live scripts contain output and graphics with the code that produced them, together in a single interactive environment called the Live functions provide additional flexibility, allowing you to pass input values and return output values.
- You can add formatted text, images, hyperlinks, and equations to live scripts and functions to produce an interactive narrative that can be shared with others.

Mechatronic System Simulation
MATLAB Live Script Example

Exercise 1.4_d

```
% Create Matrix
matrixrand=rand(5,5)
```

```
matrixrand = 5×5
    0.7577    0.7060    0.8235    0.4387    0.4898
    0.7431    0.0318    0.6948    0.3816    0.4456
    0.3922    0.2769    0.3171    0.7655    0.6463
    0.6555    0.0462    0.9502    0.7952    0.7094
    0.1712    0.0971    0.0344    0.1869    0.7547
```

```
matrixcopy = matrixrand
```

```
matrixcopy = 5×5
    0.7577    0.7060    0.8235    0.4387    0.4898
    0.7431    0.0318    0.6948    0.3816    0.4456
    0.3922    0.2769    0.3171    0.7655    0.6463
    0.6555    0.0462    0.9502    0.7952    0.7094
    0.1712    0.0971    0.0344    0.1869    0.7547
```

```
% reshape matrix
A=reshape(matrixcopy,1,25)
```

```
A = 1×25
    0.7577    0.7431    0.3922    0.6555    0.1712    0.7060    0.0318    0.2769 ···
```

```
% sort matrix
A=  sort(A);
Amin=min(reshape(matrixcopy,1,25))
```

```
Amin = 0.0318
```

```
Amax=max(reshape(A,1,25))
```

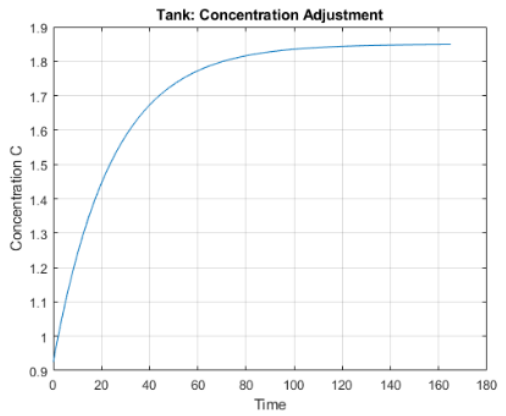
```
Amax = 0.9502
```

```
%reshape matrix A
A=reshape(A,5,5)
```

```
A = 5×5
    0.0318    0.1869    0.4387    0.6948    0.7577
    0.0344    0.2769    0.4456    0.7060    0.7655
    0.0462    0.3171    0.4898    0.7094    0.7952
    0.0971    0.3816    0.6463    0.7431    0.8235
    0.1712    0.3922    0.6555    0.7547    0.9502
```

- Save results from a Live Script including code as pdf, html or LaTex.
- Use the text editor for document your script.
- Include task description, picture and formulas.

```
21
22
23
24
25
26
27
28
29  % Plot Results
30  plot(X1(:,2), X1(:,1));
31  grid on;
32  title('Tank: Concentration Adjustment');
33  ylabel('Concentration C');
34  xlabel('Time');
```



Mechatronic System Simulation
MATLAB: Programming Hints

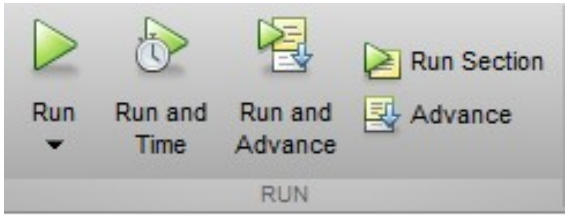
Matlab Scripts

- Script can be splitted into sub sections.

Untitled2* | Example_MSript.m*

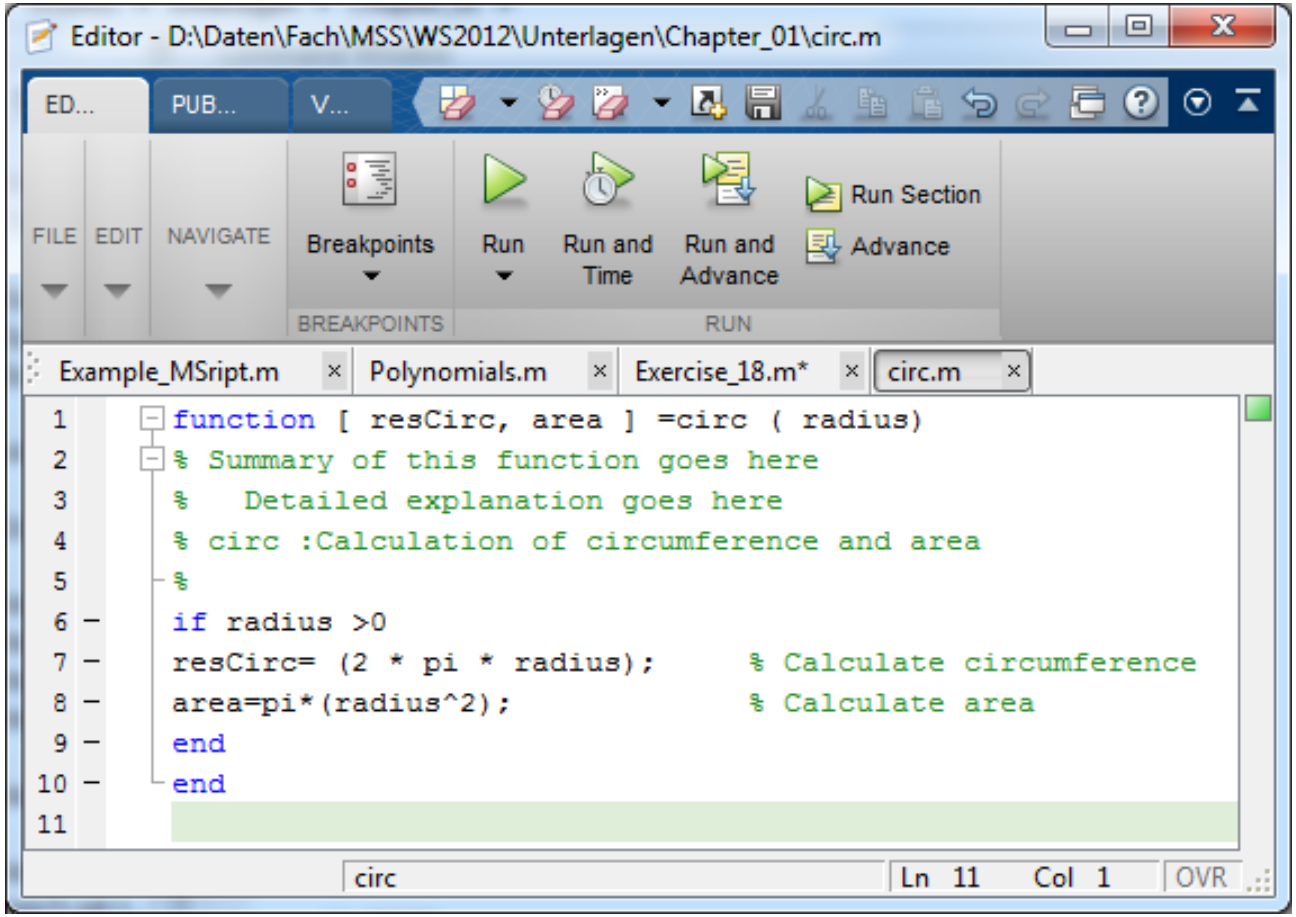
```
1   %% Example Script
2   % Exercise 1.4_d
3 - low = 20;
4 - high=100;
5
6   % Create Matrix
7 - matrixcopy = ((high-low)*rand(5,5)+low)
8
9   % reshape matrix
10- A=reshape(matrixcopy,1,25);
11
12  % sort matrix
13- A=  sort(A);
14
15  %reshape matrix A
16- A=reshape(A,5,5)
17
18  %% Subsection 1
19  % Define Variables
20
21- R=100        % Ohm
22- L=10e-3      % Henry
23- C=220e-6     % Farad
24
25
```

- Script can executed complete or in subsections.



Mechatronic System Simulation
MATLAB Script: Selection Statements

If *condition*
action
end



Mechatronic System Simulation
MATLAB Script: Selection Statements

If *condition*

action1

else

action2

end

Example_MSript.m | Polynomials.m | Exercise_18.m* | circ.m

```
1   function [ resCirc, area ] =circ ( radius)
2   % Summary of this function goes here
3   %   Detailed explanation goes here
4   % circ :Calculation of circumference and area
5   %
6   if radius >0
7       resCirc= (2 * pi * radius);     % Calculate circumference
8       area=pi*(radius^2);             % Calculate area
9       fprintf(' For a circle with the radius of %.2f,',radius)
10      fprintf(' the circumference is %.2f and area is %.2f\n',resCirc, area)
11
12  else
13      disp('Radius must be greater than 0')
14  end
15  end
```

Command Window

```
>> circ(3)
 For a circle with the radius of 3.00, the circumference is 18.85 and area is 28.27

ans =

   18.8496

>> circ(-3)
Radius must be greater than 0
>>
```

OVR

Mechatronic System Simulation
MATLAB Script: Looping

Task: Create a vector which contains the element of sqrt(x(i))

for *loopvariable* **=** *range*
action
end

1:10
1:1:10 (default step: 1)
Matlab syntax

```
1   %%For Loop
2   x=(1:10);
3   vector=[];
4   for i = [1:length(x)]
5       vector(i)=sqrt (x(i));
6   end
7
```

while *condition*
action
end

```
9   %% While Loop
10  x= (1:10);
11  vector=[]
12  i=1;
13  while i <= length(x)
14      vector(i)=sqrt(x(i));
15      i=i+1;
16  end
17
```

Ln 10 Col 1 OVR

Hint: „Ctrl +C" stops Script execution!

Mechatronic System Simulation
MATLAB as a Programming Languages

Flow Control Structures in MATLAB

`break`	break of a loop
`end`	end of a loop or an `if`-structure
`error`	output an error message and stop execution
`for`	for loop (counting)
`if, else, elseif`	conditional program flow
`return`	return to the calling function
`switch, case, otherwise`	decision between several cases
`while`	while loop (condition)

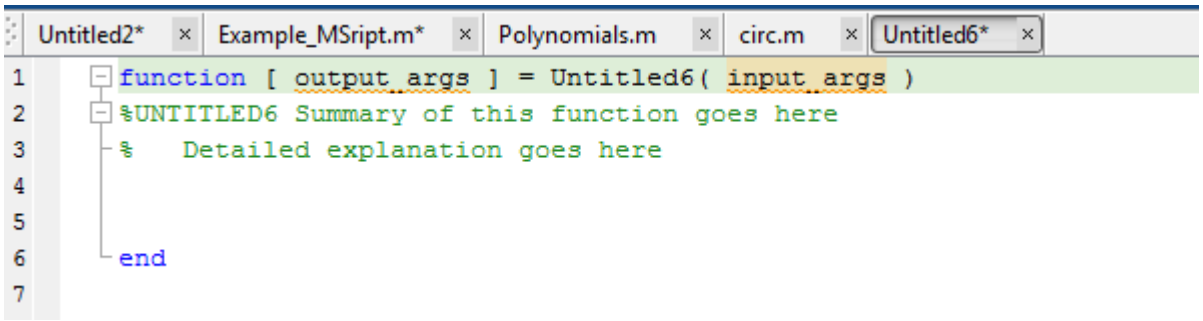
Mechatronic System Simulation
MATLAB as a Programming Language

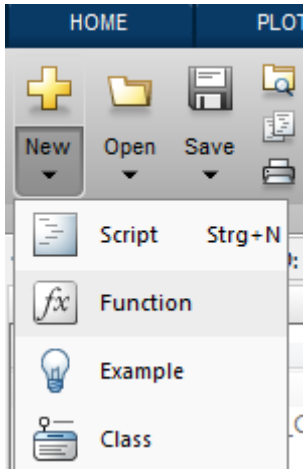
MATLAB Scripts function/ Live Script function: User Defined Functions

Definition: function [output] = functionname (input)

Input parameter (input into function)
Output parameter (output from function)
(Local) variables are not visible in Workspace!

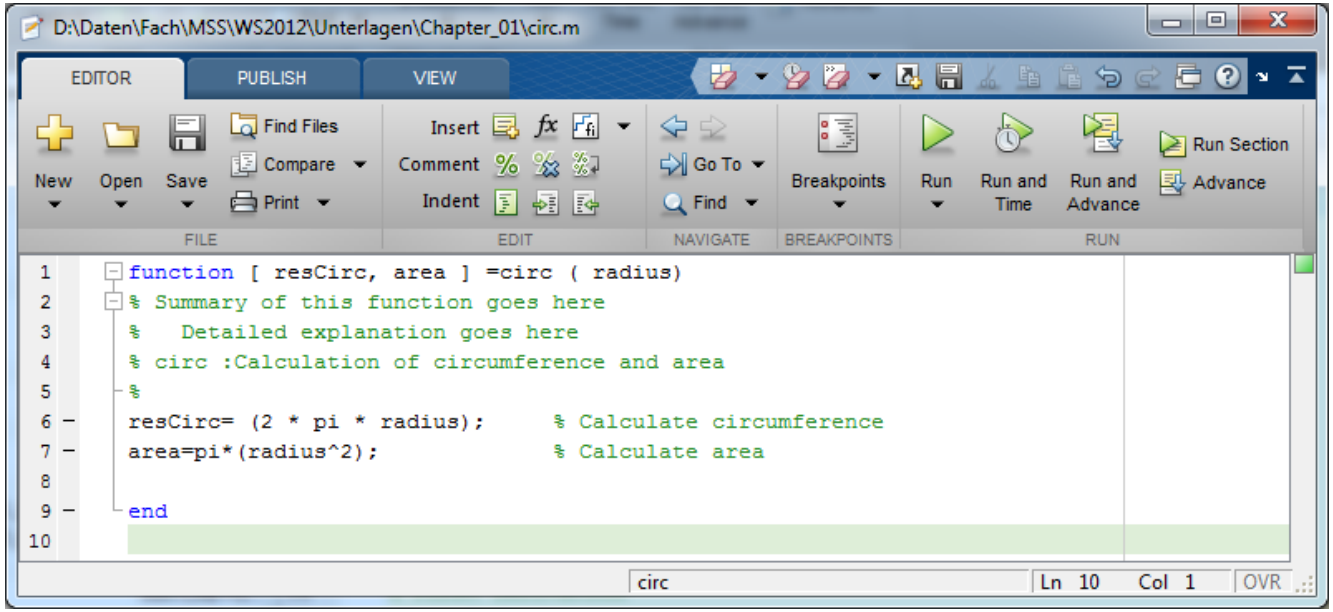
- Function needs to be defined as a *function*.
- Function Script will be saved as a Script





Mechatronic System Simulation
MATLAB Script Function (User Defined)

Task: „Calculate circumference and area of a circle“
Input: radius
Output: circumference, circle area





Mechatronic System Simulation
MATLAB Graphics

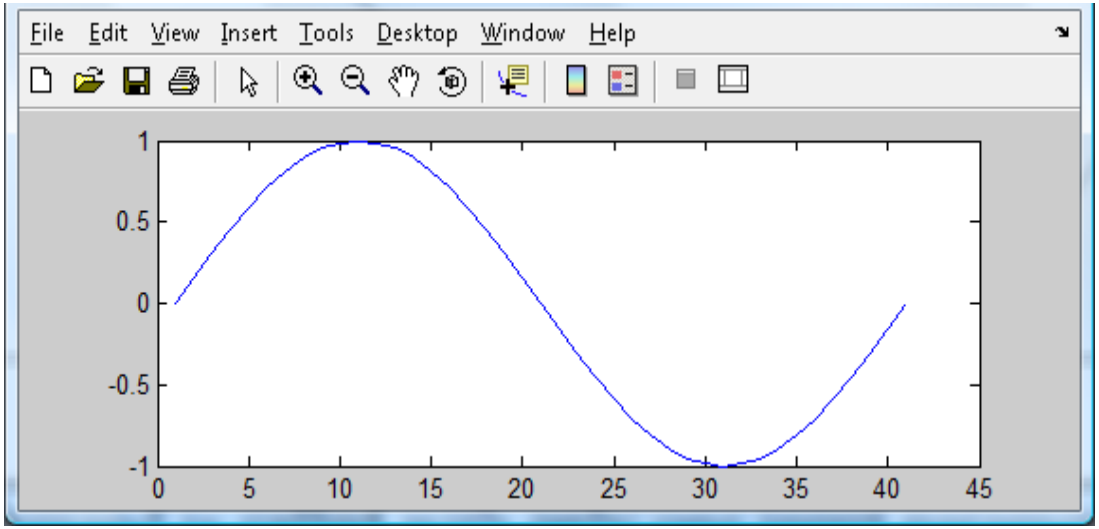
Plot of Vectors and Matrices: 2 D Graphic

Example: Harmonic signals (y versus samples)

```
%% Plot y versus index
%  u(t)= U * sin (x(t))
U=1;                    % Define variable U=1
x=0:pi/20:2*pi;         % Define Vector X from 0 to 2*pi in steps of pi/20
u = U * sin (x);        % Calculate results for u

plot (u);               % Create figure from x versus index
```

plot(u): If the plot–command is called with only one input parameter, the vector is plotted versus its indices.



Mechatronic System Simulation
MATLAB Graphics

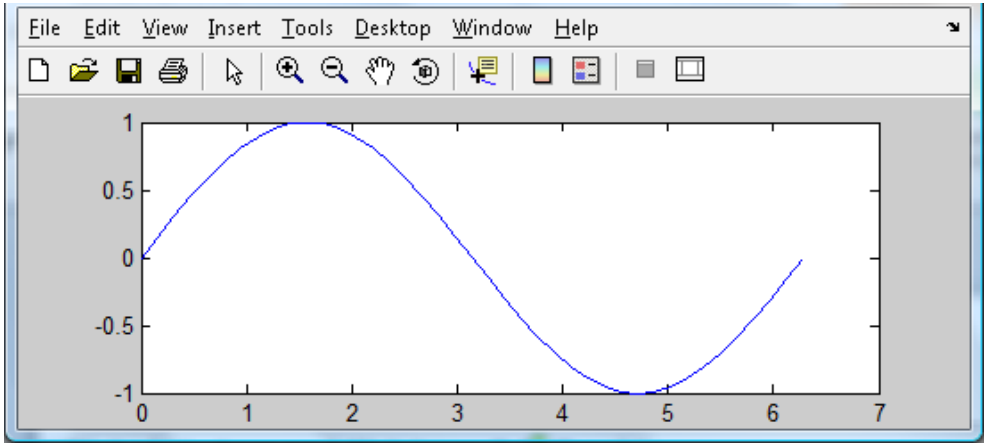
Plot of Vectors and Matrices: 2 D Graphic

Example: Harmonic signals (y versus x)

```
%% Plot x versus y
% % u(t)= U * sin (x(t))

U=1;                    % Define variable U=1
x=0:pi/20:2*pi;         % Define Vector X from 0 to 2*pi in steps of pi/20
u = U * sin (x);        % Calculate results for u
plot (x,u)              % Create figure from x and u
```

plot(x,u): Using two vectors as input parameters the second vector is plotted versus the first vector.



Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices: 2 D Graphic

Example: Plot of two harmonic signals in one figure

```
%
% u(t)= U * sin (x(t))

U=1;                    % Define variable U=1
x=0:pi/20:2*pi;         % Define Vector X from 0 to 2*pi in steps of pi/20
u = U * sin (x);        % Calculate results for u
u1 = U*sin (x+pi/4);    % Calculate results for u1

plot (x, u, x, u1)      % Plot signals

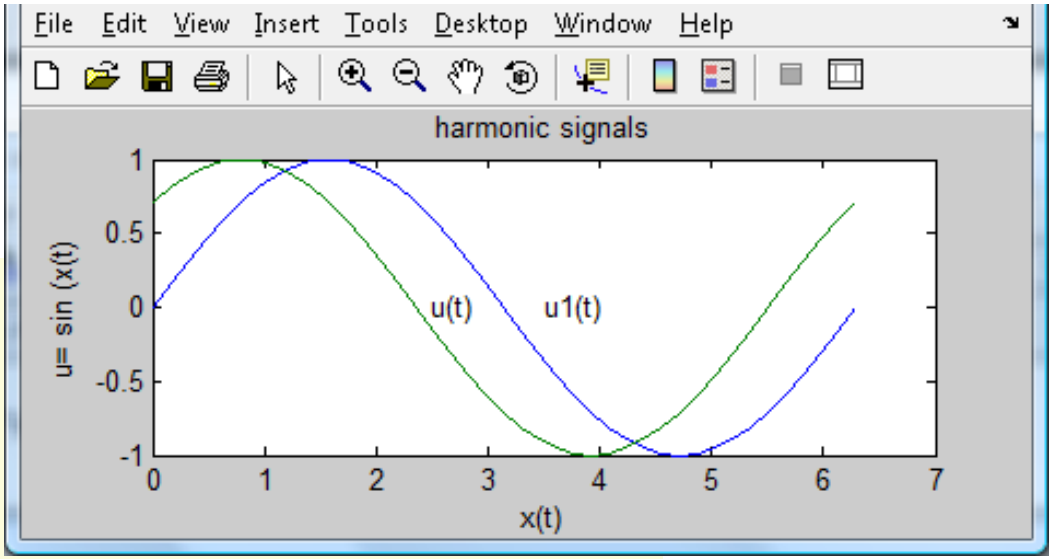
plot(x, [u; u1;u2])     % 1 alternative command

plot (x,u);             % 2. alternative commands: create plot with u
hold on                 % hold acitve chart for add data
plot (x,u1)             % add data u1 to figure
hold off                % release figure
```



Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices: 2 D Graphic: Title, labels, text




```
U=1;
x=0:pi/20:2*pi;
u = U * sin (x);
u1 = U*sin (x+pi/4);

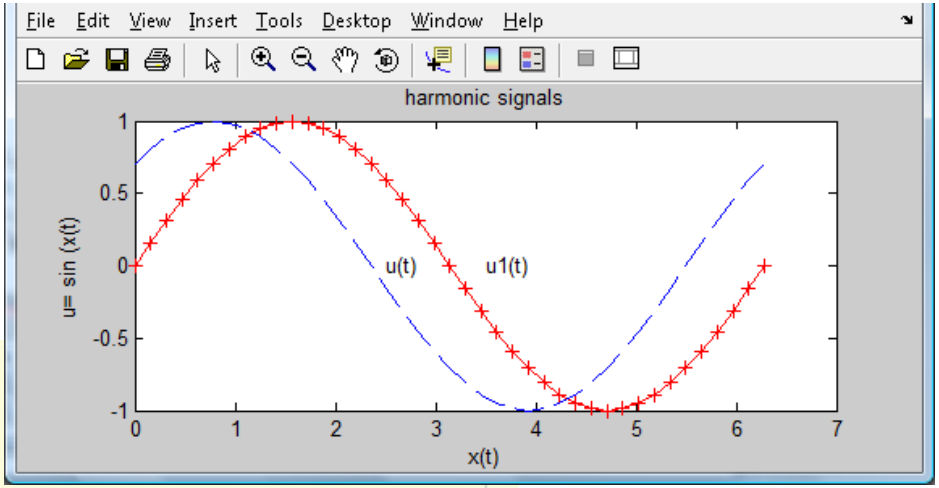
plot (x, u)
hold on;
plot (x,u1)
hold off

title ('harmonic signals')      % adds text at the top
xlabel ('x(t)')                 % adds text beside the X- axis
ylabel (' u= sin (x(t) ')       % adds text beside the Y axis
text (2.5,0, 'u(t)')            % adds the lable in the chart
text (3.5,0, 'u1(t)')           % adds the lable in the chart
```

Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices: 2 D Graphic:

Colours, Markers, Line Types :




```
U=1;
X=0:pi/20:2*pi;
u = U * sin (x);
u1 = U*sin (x+pi/4);

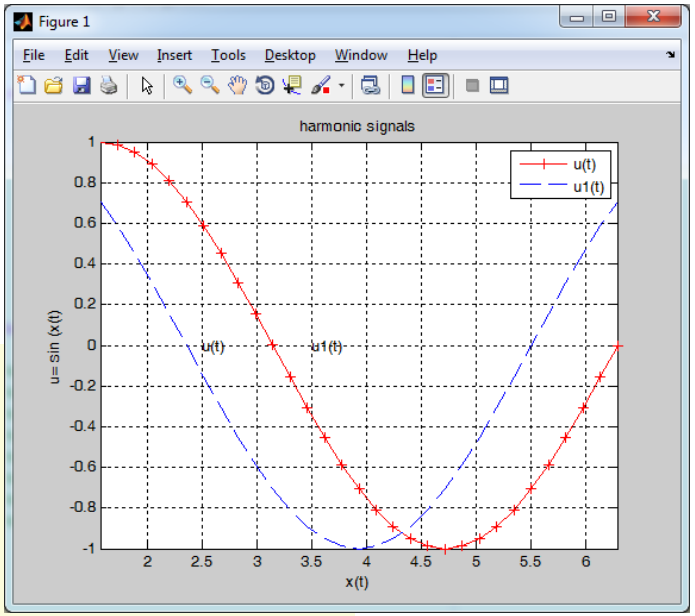
plot (x, u, 'r+-')           % plot line in red, plus sign and dashed style
hold on;
plot ( x, u1,  'b--' )       % plot line in blue and dashed style
hold off;

title ('harmonic signals')   % adds text at the top
xlabel ('x(t)')              % adds text beside the X- axis
ylabel (' u= sin (x(t); u1= sin (x+pi/4) ')  % adds text beside the Y axis
text (2.5,0, 'u(t)')         % adds the lable in the chart
text (3.5,0, 'u1(t)')        % adds the lable in the chart
```

Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices: 2 D Graphic:

Axis ranges, grids, legend




```
U=1;
x=0:pi/20:2*pi;
u = U * sin (x);
u1 = U*sin (x+pi/4);
plot (x, u, 'r+-', x, u1,  'b--' )

axis ([pi/2 2*pi -1 1]);     % axis = [x1 x2 y1 y2] (range plot)
grid on                      % display grid
legend ('u(t)', 'u1(t)')     % generates a legend
title ('harmonic signals')   % adds text at the top
xlabel ('x(t)')              % adds text beside the X- axis
ylabel (' u= sin (x(t) ')    % adds text beside the Y axis
text (2.5,0, 'u(t)')         % adds the lable in the chart
text (3.5,0, 'u1(t)')        % adds the lable in the chart
```

Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices: 2 D Graphic:

Symbol	Color	Symbol	Marker	Symbol	Line Type
y	yellow	.	dots	-	solid
m	magenta	o	circle	:	dotted
c	cyan	x	cross	-.	dash-dot
r	red	+	plus sign	--	dashed
g	green	*	asterisk		
b	blue	s	square		
w	white	d	diamond		
k	black	p	pentagram		

```
>> plot (x, u, 'r+-', x, u1, 'b--' )
>> axis                                % scale the axis
>> grid on grid off                    % Switch grid on/off
>> legend                              % generates a legend
```


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Mechatronic System Simulation
MATLAB Graphics

Plot of Vectors and Matrices in several graphics:

subplot(m,n,nr) breaks the figure in m by n subplots and activate the the current plot (counted rowwise)

```
U=1;
x=0:pi/20:2*pi;
u = U * sin (x);
u1 = U*sin (x+pi/4);

subplot(2,2,1)          % split figure in 2 rows and 2 column,
                        % activate suplot 1
stem (x, u,'dr' )       % Plot discrete sequence data
title ('u(t) = U* sin(t)')
xlabel('x(t)')
ylabel('u= sin (x(t) ')
grid on;
axis ([pi/2 2*pi -1.3 1.3]);

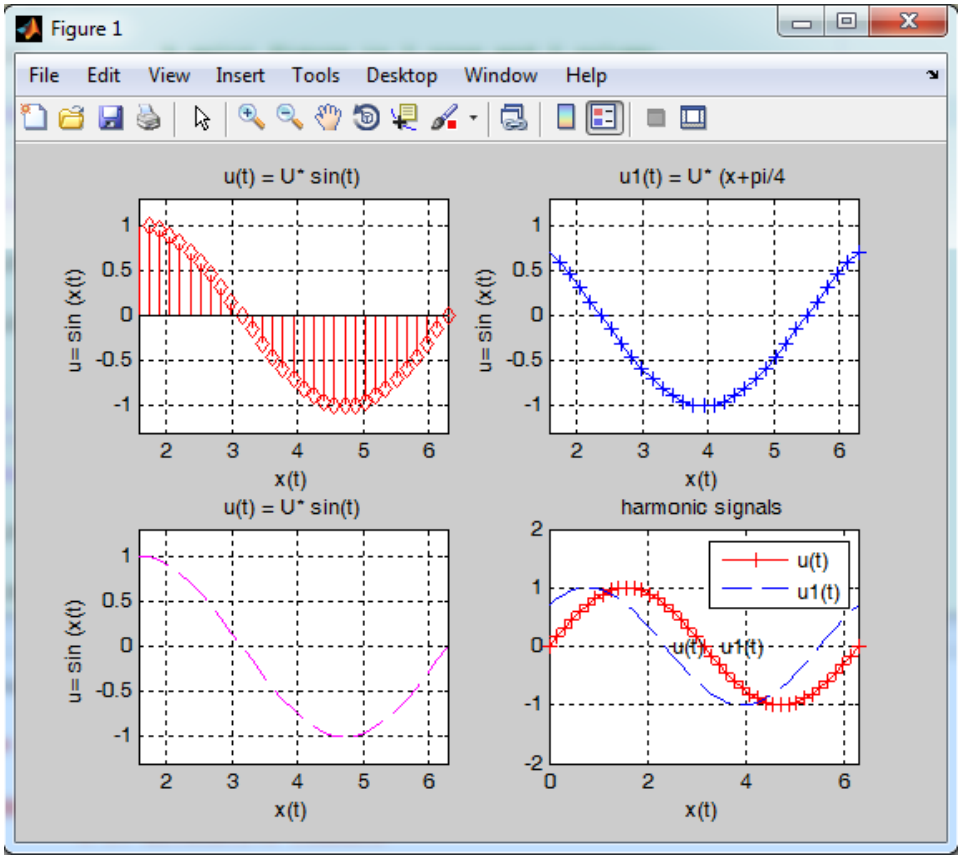
subplot(2,2,2)          % split figure in 2 rows and 2 column,
                        % actvate suplot 2
plot (x, u1,  'b+-' )
title ('u1(t) = U* (x+pi/4');
xlabel('x(t)')
ylabel('u= sin (x(t) ')
grid on;
axis ([pi/2 2*pi -1.3 1.3]);

subplot(2,2,3)          % split figure in 2 rows and 2 column,
                        % actvate suplot 3
plot (x, u,  'm--' )
title ('u(t) = U* sin(t)')
xlabel('x(t)')
ylabel('u= sin (x(t) ')
grid on;
axis ([pi/2 2*pi -1.3 1.3]);

subplot(2,2,4)          % split figure in 2 rows and 2 column,
                        % actvate suplot 1

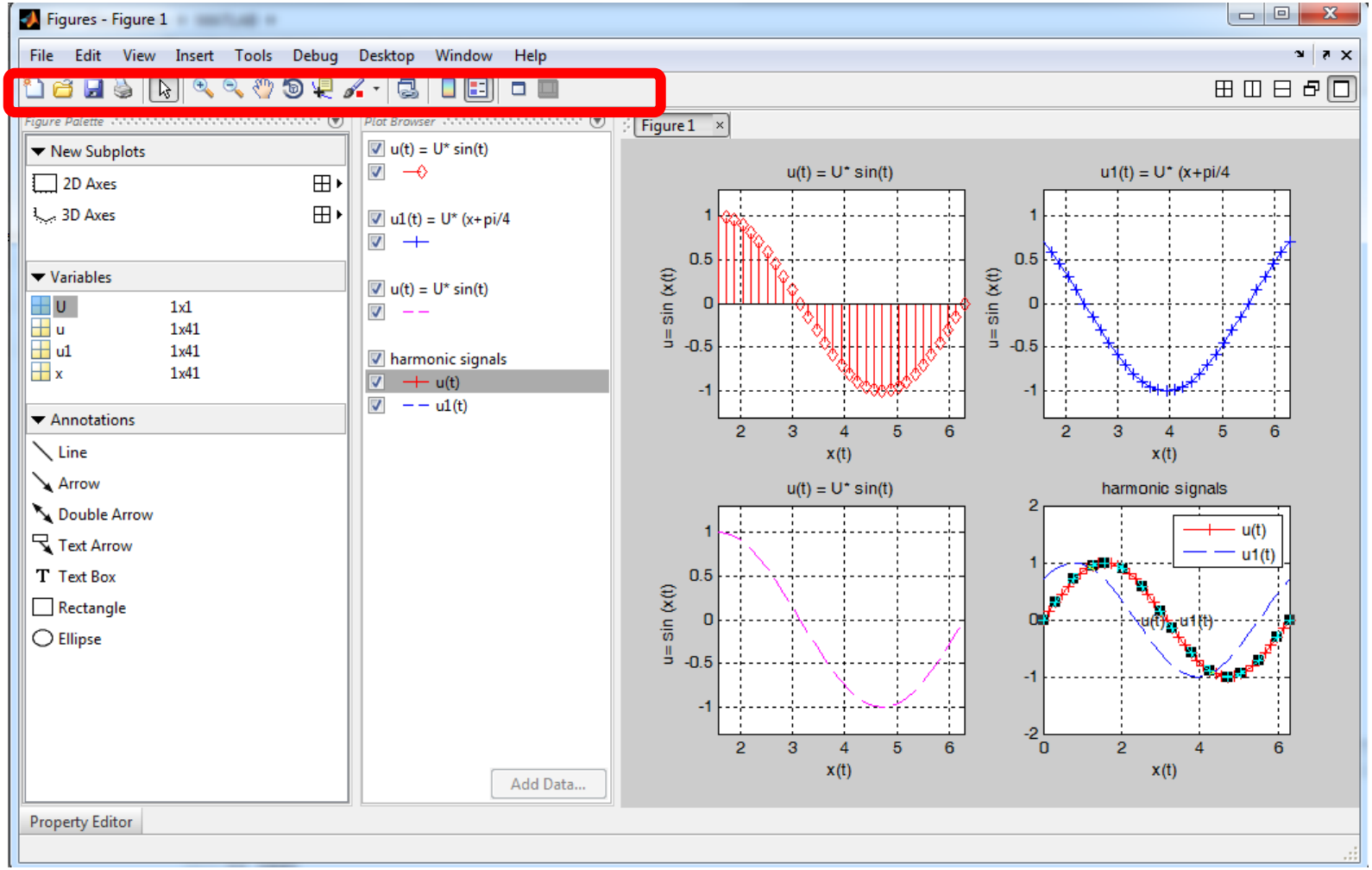
plot (x, u, 'r+-', x, u1,  'b--' )

axis ([0 2*pi -2 2]);
grid on
legend ('u(t)', 'u1(t)')
title ('harmonic signals')
xlabel ('x(t)')
text (2.5,0, 'u(t)')
text (3.5,0, 'u1(t)')
```



Mechatronic System Simulation
MATLAB Graphics

Post Procesing tools for graphics



Exercises

1.5 – 1.8

Mechatronic System Simulation

Seminar Overview: Contents

Part 1: Code based models

- Introduction
- MATLAB interactive
- MATLAB as a programming language
- MATLAB Graphics
- MATLAB functions (built in)
- MATLAB Data Interfaces (In-/ Output)

Mechatronic System Simulation
MATLAB: Polynomials function (roots, derivation)

Matlab represents a polynomials as a row vector of coefficients.

$$f(x) = x^3 + 2x^2 - 4x + 3$$

Degree:	3
Vector with coefficient:	p=[1 2 -4 3] * Non-exiting polynomials must be specified with 0.
Roots(f(x) = 0):	f0=roots(p)
Evaluate polynom at x= 3:	fvalue=polyval(p,3)
Evaluate polynom at x=1 to 5:	fvalue= polyval(p,1:5)
Derivate Polynom:	f1=polyder(p)

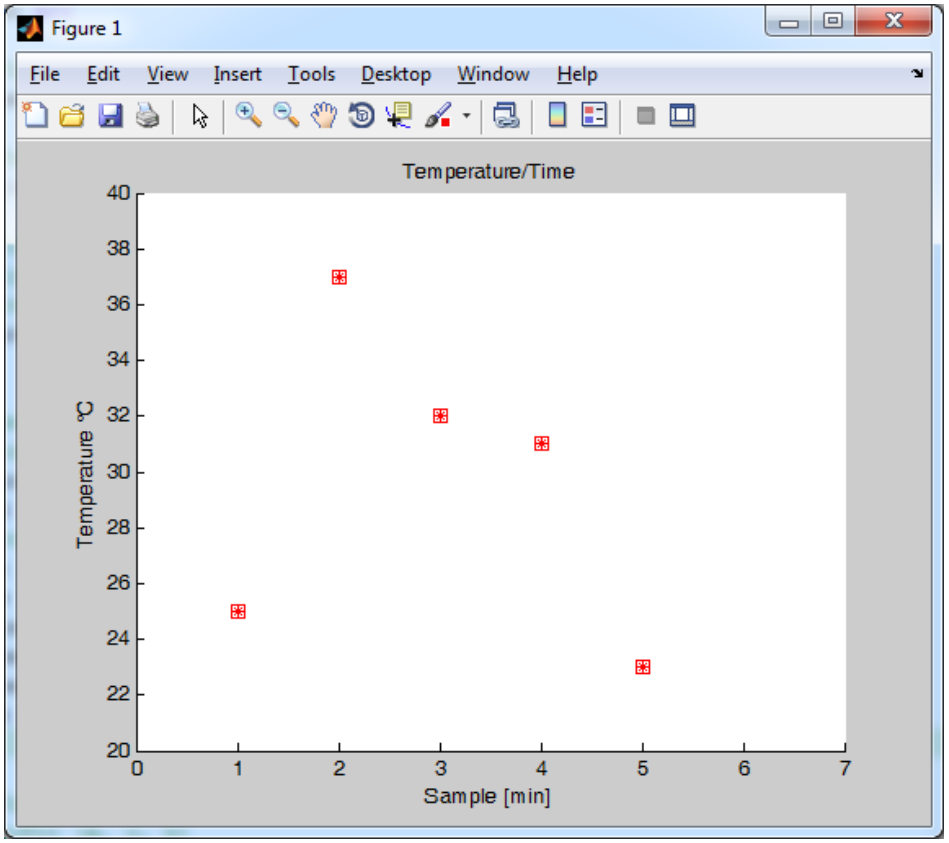
Mechatronic System Simulation
MATLAB: Polynomial Functions

poly	characteristic polynomial	p=poly(A) p = 1.0000 1.0000 1.0000
roots	roots of the polynomial	r=roots(p) r = -0.5000 + 0.8660i -0.5000 - 0.8660i
polyval	evaluate polynomial	GZ=polyval(p, x);
polyvalm	evaluate polynomial with matrix argument	
conv	polynomial multiplication	p=conv([1 -r(1)], [1 -r(2)]) p = 1.0000 1.0000 1.0000
deconv	polynomial division	deconv(p ,[1 -r(2)]) ans = 1.0000 0.5000-0.8660i
residue	partial-fraction expansion	residue(b,a) ans = 0 - 1.1547i 0 + 1.1547i
polyder	differentiate polynomial	polyder(a) ans = 2 1
polyfit	fit polynomial to data in a least-squares sense	p=polyfit(a,p,3) p = 1.0000 0 0 0

Mechatronic System Simulation
MATLAB: Curve Fitting function(interpolation)

Example: Data samples

	A	B
1	Sample	Temperature
2	Min	°C
3	1	25
4	2	27
5	3	32
6	4	31
7	5	23



Mechatronic System Simulation
MATLAB: Curve Fitting function(interpolation)

Task: Estimate values of a curve between the samples (e.g. between 2:30 min).

Concept of solution:

- Extrapolate (Estimate outside the bounds of recorded data).
- Interpolate (Estimate the values between the recorded data points.
- Curve Fitting (Finding the curve that „best fits" the data).

Matlab function *polyfit* combines both methods. The function finds the coefficient of polynomial of a specified degree that best fits the data.

Mechatronic System Simulation
MATLAB: Curve Fitting function(interpolation)

Grade =1 ☹

Polyfit.m*

```
16      %Define vector values
17 -    temperature =[25 27 32 31 23]
18
19      %Define vector t
20 -    sample=[1:5]
21
22      %Define Grade =1 for Polyfit function
23      %polyfit (x, y, n)
24      %calculation of polynominal data
25 -    coeficient= polyfit (sample,temperature,1)
26
27      %Calculate values for chart
28 -    curve= polyval (coeficient,sample)
29
30      %Draw chart
31 -    plot (sample,temperature,'r*');          %plot the value over t
32 -    hold on                                  %hold values in graph and add an addional one
33 -    plot (sample,curve)                      %plot the polynominal
34 -    axis ([0 7 20 40])                       %Scale x (0- 7) and y (20- 40)
35 -    title('Temperature/Time')
36 -    xlabel('Sample [min]')
37 -    ylabel('Temperature °C')
38 -    hold off
39
40
```

Command Window

```
temperature =

    25    27    32    31    23

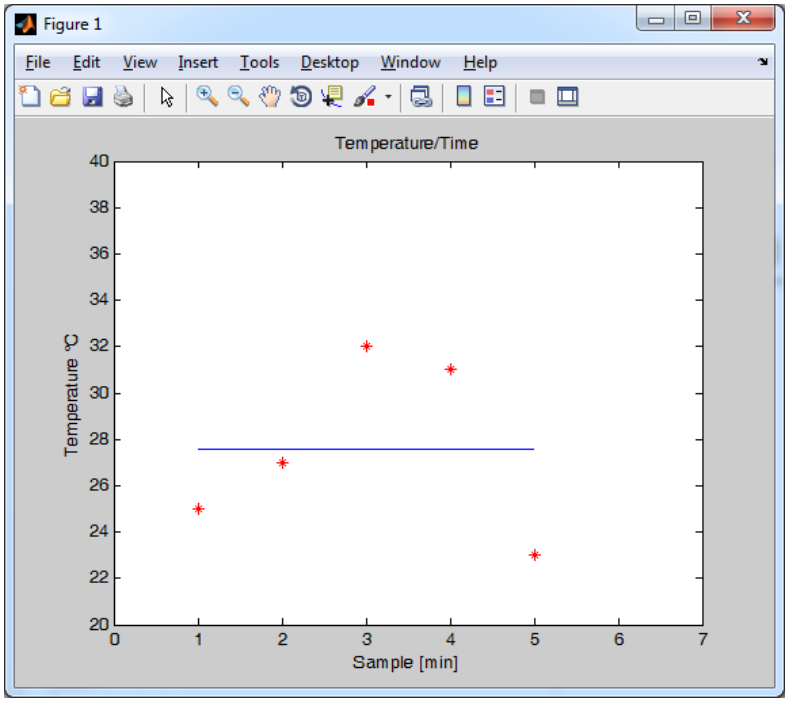

sample =

     1     2     3     4     5


coeficient =

    0.0000   27.6000

fx >>
```



Mechatronic System Simulation
MATLAB: Curve Fitting function(interpolation)

Grade =4 ☺

Variables - Matrix Help

Polyfit.m*

```
15
16      %Define vector values
17 -    temperature =[25 27 32 31 23]
18
19      %Define vector t
20 -    sample=[1:5]
21
22      %Define Grade =4 for Polyfit function
23      %polyfit (x, y, n)
24      %calculation of polynominal data
25 -    coeficient= polyfit (sample,temperature,4)
26
27      %Calculate values for chart
28 -    curve= polyval (coeficient,sample)
29
30      %Draw chart
31 -    plot (sample,temperature,'r*');
32 -    hold on
33 -    plot (sample,curve)
34 -    axis ([0 7 20 40])
35 -    title('Temperature/Time')
36 -    xlabel('Sample [min]')
37 -    ylabel('Temperature °C')
38 -    hold off
```

Command Window

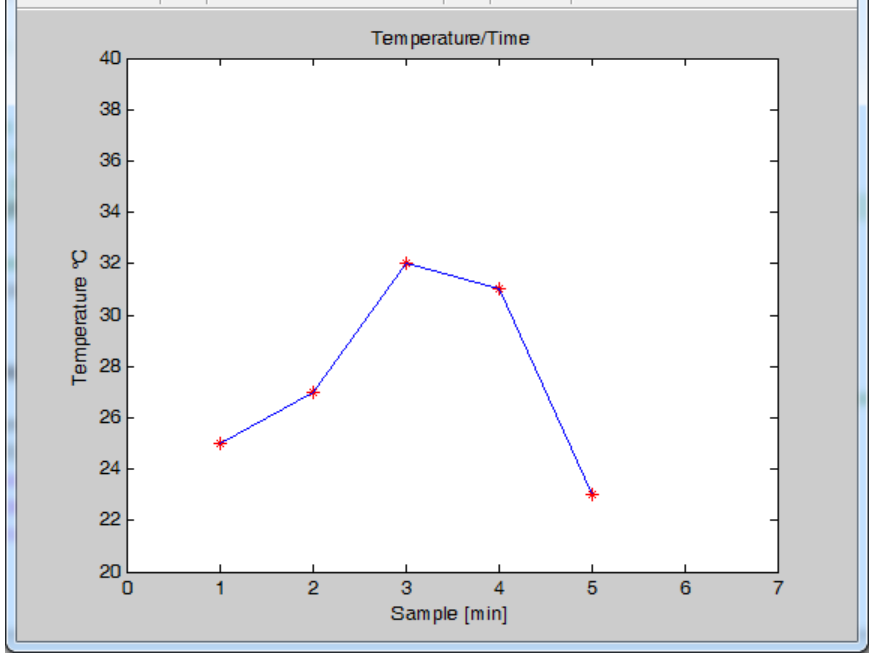
```
coeficient =

    0.3333   -4.8333   22.1667  -35.6667   43.0000


curve =

   25.0000   27.0000   32.0000   31.0000   23.0000

fx >>
```



Mechatronic System Simulation
MATLAB Data Interfaces (In-/ Output)

Reading data filtered by numeric values ,text and raw data.

	A	B	C	D	E	F
1	Pos	Value1	Value2	Value3	Value4	Comment
2	1	0,340386	0,505957	0,138624	0,814285	OK
3	2	0,585268	0,699077	0,149294	0,243525	OK
4	3	0,223812	0,890903	0,257508	0,929264	Not OK
5	4	0,751267	0,959291	0,840717	0,349984	Not OK
6	5	0,255095	0,547216	0,254282	0,196595	OK
7						
8						

Workspace

Name	Size	Value	Min
numbers	5x5	<5x5 double>	0.13
rawdata	6x6	<6x6 cell>	
text	6x6	<6x6 cell>	

Command Window

```
>> [numbers, text, rawdata]=xlsread('matrix1.xls')

numbers =

    1.0000    0.3404    0.5060    0.1386    0.8143
    2.0000    0.5853    0.6991    0.1493    0.2435
    3.0000    0.2238    0.8909    0.2575    0.9293
    4.0000    0.7513    0.9593    0.8407    0.3500
    5.0000    0.2551    0.5472    0.2543    0.1966


text =

    'Pos'    'Value1'    'Value2'    'Value3'    'Value4'    'Comment'
    ''       ''          ''          ''          ''          'OK'
    ''       ''          ''          ''          ''          'OK'
    ''       ''          ''          ''          ''          'Not OK'
    ''       ''          ''          ''          ''          'Not OK'
    ''       ''          ''          ''          ''          'OK'


rawdata =

    'Pos'    'Value1'    'Value2'    'Value3'    'Value4'    'Comment'
    [  1]    [0.3404]    [0.5060]    [0.1386]    [0.8143]    'OK'
    [  2]    [0.5853]    [0.6991]    [0.1493]    [0.2435]    'OK'
    [  3]    [0.2238]    [0.8909]    [0.2575]    [0.9293]    'Not OK'
    [  4]    [0.7513]    [0.9593]    [0.8407]    [0.3500]    'Not OK'
    [  5]    [0.2551]    [0.5472]    [0.2543]    [0.1966]    'OK'

fx >>
```

xlsread function always returns the numbers first, second the text and third the raw data.

Mechatronic System Simulation
MATLAB Data Interfaces (In-/ Output)

Write data to file (Spreadsheet)

Command Window

```
>> xlswrite('matrix3.xlsx',Matrix)
>>
```

matrix | nums | txt | subset | Matrix

Matrix <5x5 double>

	1	2	3	4	5	6
1	1	0.3404	0.5060	0.1386	0.8143	
2	2	0.5853	0.6991	0.1493	0.2435	
3	3	0.2238	0.8909	0.2575	0.9293	
4	4	0.7513	0.9593	0.8407	0.3500	
5	5	0.2551	0.5472	0.2543	0.1966	
6						

	A	B	C	D	E
1	1	0,340386	0,505957	0,138624	0,814285
2	2	0,585268	0,699077	0,149294	0,243525
3	3	0,223812	0,890903	0,257508	0,929264
4	4	0,751267	0,959291	0,840717	0,349984
5	5	0,255095	0,547216	0,254282	0,196595
6					